

Sección I. Integrales

TABLA

1

TABLA DE INTEGRALES INDEFINIDAS

En estas tablas de integrales indefinidas (primitivas), a, b, p, q, n son constantes, restringidas si así se especifica; $e = 2,71828\dots$ es la base de los logaritmos naturales. $\ln u$ denota el logaritmo natural de u , donde se sobreentiende que $u > 0$ [en general, para extender la validez de una fórmula al caso $u < 0$ basta sustituir $\ln u$ por $\ln |u|$]. Todos los ángulos están medidos en radianes. Se han omitido todas las constantes de integración. Se supone además que la división por cero está excluida.

Hemos clasificado las integrales por tipos, según cuáles de estas expresiones o funciones están contenidas en el integrando:

- | | | |
|---------------------------------------|--|--|
| (1) $ax + b$ | (13) $\sqrt{ax^2 + bx + c}$ | (25) e^{ax} |
| (2) $\sqrt{ax + b}$ | (14) $x^3 + a^3$ | (26) $\ln x$ |
| (3) $ax + b$ y $px + q$ | (15) $x^4 \pm a^4$ | (27) $\operatorname{sh} ax$ |
| (4) $\sqrt{ax + b}$ y $px + q$ | (16) $x^n \pm a^n$ | (28) $\operatorname{ch} ax$ |
| (5) $\sqrt{ax + b}$ y $\sqrt{px + q}$ | (17) $\operatorname{sen} ax$ | (29) $\operatorname{sh} ax$ y $\operatorname{ch} ax$ |
| (6) $x^2 + a^2$ | (18) $\cos ax$ | (30) $\operatorname{th} ax$ |
| (7) $x^2 - a^2$, con $x^2 > a^2$ | (19) $\operatorname{sen} ax$ y $\cos ax$ | (31) $\operatorname{coth} ax$ |
| (8) $a^2 - x^2$, con $x^2 < a^2$ | (20) $\operatorname{tg} ax$ | (32) $\operatorname{sech} ax$ |
| (9) $\sqrt{x^2 + a^2}$ | (21) $\cot ax$ | (33) $\operatorname{csch} ax$ |
| (10) $\sqrt{x^2 - a^2}$ | (22) $\sec ax$ | (34) funciones hiperbólicas inversas |
| (11) $\sqrt{a^2 - x^2}$ | (23) $\csc ax$ | |
| (12) $ax^2 + bx + c$ | (24) funciones trigonométricas inversas | |

En algunas integrales aparecen los números B_n de Bernoulli y los E_n de Euler, definidos en las páginas 135-136.

(1) INTEGRALES QUE CONTIENEN $ax + b$

$$1.1.1 \quad \int \frac{dx}{ax + b} = \frac{1}{a} \ln(ax + b)$$

$$1.1.2 \quad \int \frac{x dx}{ax + b} = \frac{x}{a} - \frac{b}{a^2} \ln(ax + b)$$

$$1.1.3 \quad \int \frac{x^2 dx}{ax + b} = \frac{(ax + b)^2}{2a^3} - \frac{2b(ax + b)}{a^3} + \frac{b^2}{a^3} \ln(ax + b)$$

$$1.1.4 \quad \int \frac{dx}{x(ax + b)} = \frac{1}{b} \ln\left(\frac{x}{ax + b}\right)$$

$$1.1.5 \quad \int \frac{dx}{x^2(ax + b)} = -\frac{1}{bx} + \frac{a}{b^2} \ln\left(\frac{ax + b}{x}\right)$$

$$1.1.6 \quad \int \frac{dx}{(ax+b)^2} = \frac{-1}{a(ax+b)}$$

$$1.1.7 \quad \int \frac{x dx}{(ax+b)^2} = \frac{b}{a^2(ax+b)} + \frac{1}{a^2} \ln(ax+b)$$

$$1.1.8 \quad \int \frac{x^2 dx}{(ax+b)^2} = \frac{ax+b}{a^3} - \frac{b^2}{a^3(ax+b)} - \frac{2b}{a^3} \ln(ax+b)$$

$$1.1.9 \quad \int \frac{dx}{x(ax+b)^2} = \frac{1}{b(ax+b)} + \frac{1}{b^2} \ln\left(\frac{x}{ax+b}\right)$$

$$1.1.10 \quad \int \frac{dx}{x^2(ax+b)^2} = \frac{-a}{b^2(ax+b)} - \frac{1}{b^2x} + \frac{2a}{b^3} \ln\left(\frac{ax+b}{x}\right)$$

$$1.1.11 \quad \int \frac{dx}{(ax+b)^3} = \frac{-1}{2(ax+b)^2}$$

$$1.1.12 \quad \int \frac{x dx}{(ax+b)^3} = \frac{-1}{a^2(ax+b)} + \frac{b}{2a^2(ax+b)^2}$$

$$1.1.13 \quad \int \frac{x^2 dx}{(ax+b)^3} = \frac{2b}{a^3(ax+b)} - \frac{b^2}{2a^3(ax+b)^2} + \frac{1}{a^3} \ln(ax+b)$$

$$1.1.14 \quad \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)a}. \quad \text{Si } n = -1, \text{ véase 1.1.1.}$$

$$1.1.15 \quad \int x(ax+b)^n dx = \frac{(ax+b)^{n+2}}{(n+2)a^2} - \frac{b(ax+b)^{n+1}}{(n+1)a^2}, \quad n \neq -1, -2$$

Si $n = -1, -2$, véanse 1.1.2 y 1.1.7.

$$1.1.16 \quad \int x^2(ax+b)^n dx = \frac{(ax+b)^{n+3}}{(n+3)a^3} - \frac{2b(ax+b)^{n+2}}{(n+2)a^3} + \frac{b^2(ax+b)^{n+1}}{(n+1)a^3}$$

Si $n = -1, -2, -3$, véanse 1.1.3, 1.1.8 y 1.1.13.

$$1.1.17 \quad \int x^m(ax+b)^n dx = \begin{cases} \frac{x^{m+1}(ax+b)^n}{m+n+1} + \frac{nb}{m+n+1} \int x^m(ax+b)^{n-1} dx \\ \frac{x^m(ax+b)^{n+1}}{(m+n+1)a} - \frac{mb}{(m+n+1)a} \int x^{m-1}(ax+b)^n dx \\ -\frac{x^{m+1}(ax+b)^{n+1}}{(n+1)b} + \frac{m+n+2}{(n+1)b} \int x^m(ax+b)^{n+1} dx \end{cases}$$

(2)

INTEGRALES QUE CONTIENEN $\sqrt{ax+b}$

$$1.2.1 \quad \int \frac{dx}{\sqrt{ax+b}} = \frac{2\sqrt{ax+b}}{a}$$

$$1.2.2 \quad \int \frac{x dx}{\sqrt{ax+b}} = \frac{2(ax-2b)}{3a^2} \sqrt{ax+b}$$

$$1.2.3 \quad \int \frac{x^2 dx}{\sqrt{ax+b}} = \frac{2(3a^2x^2 - 4abx + 8b^2)}{15a^3} \sqrt{ax+b}$$

$$1.2.4 \quad \int \frac{dx}{x\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{b}} \ln \left(\frac{\sqrt{ax+b} - \sqrt{b}}{\sqrt{ax+b} + \sqrt{b}} \right) \\ \frac{2}{\sqrt{-b}} \operatorname{tg}^{-1} \sqrt{\frac{ax+b}{-b}} \end{cases}$$

$$1.2.5 \quad \int \frac{dx}{x^2\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b} \int \frac{dx}{x\sqrt{ax+b}} \quad [\text{Véase 1.2.12}].$$

$$1.2.6 \quad \int \sqrt{ax+b} dx = \frac{2\sqrt{(ax+b)^3}}{3a}$$

$$1.2.7 \quad \int x\sqrt{ax+b} dx = \frac{2(3ax-2b)}{15a^2} \sqrt{(ax+b)^3}$$

$$1.2.8 \quad \int x^2\sqrt{ax+b} dx = \frac{2(15a^2x^2 - 12abx + 8b^2)}{105a^3} \sqrt{(ax+b)^3}$$

$$1.2.9 \quad \int \frac{\sqrt{ax+b}}{x} dx = 2\sqrt{ax+b} + b \int \frac{dx}{x\sqrt{ax+b}} \quad [\text{Véase 1.2.12}].$$

$$1.2.10 \quad \int \frac{\sqrt{ax+b}}{x^2} dx = -\frac{\sqrt{ax+b}}{x} + \frac{a}{2} \int \frac{dx}{x\sqrt{ax+b}} \quad [\text{Véase 1.2.12}].$$

$$1.2.11 \quad \int \frac{x^m}{\sqrt{ax+b}} dx = \frac{2x^m\sqrt{ax+b}}{(2m+1)a} - \frac{2mb}{(2m+1)a} \int \frac{x^{m-1}}{\sqrt{ax+b}} dx$$

$$1.2.12 \quad \int \frac{dx}{x^m\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{(m-1)bx^{m-1}} - \frac{(2m-3)a}{(2m-2)b} \int \frac{dx}{x^{m-1}\sqrt{ax+b}}$$

$$1.2.13 \quad \int x^m\sqrt{ax+b} dx = \frac{2x^m}{(2m+3)a} (ax+b)^{3/2} - \frac{2mb}{(2m+3)a} \int x^{m-1}\sqrt{ax+b} dx$$

$$1.2.14 \quad \int \frac{\sqrt{ax+b}}{x^m} dx = -\frac{\sqrt{ax+b}}{(m-1)x^{m-1}} + \frac{a}{2(m-1)} \int \frac{dx}{x^{m-1}\sqrt{ax+b}}$$

$$1.2.15 \quad \int \frac{\sqrt{ax+b}}{x^m} dx = \frac{-(ax+b)^{3/2}}{(m-1)bx^{m-1}} - \frac{(2m-5)a}{(2m-2)b} \int \frac{\sqrt{ax+b}}{x^{m-1}} dx$$

$$1.2.16 \quad \int (ax+b)^{m/2} dx = \frac{2(ax+b)^{(m+2)/2}}{a(m+2)}$$

$$1.2.17 \quad \int x(ax+b)^{m/2} dx = \frac{2(ax+b)^{(m+4)/2}}{a^2(m+4)} - \frac{2b(ax+b)^{(m+2)/2}}{a^2(m+2)}$$

$$1.2.18 \quad \int x^2(ax+b)^{m/2} dx = \frac{2(ax+b)^{(m+6)/2}}{a^3(m+6)} - \frac{4b(ax+b)^{(m+4)/2}}{a^3(m+4)} + \frac{2b^2(ax+b)^{(m+2)/2}}{a^3(m+2)}$$

$$1.2.19 \quad \int \frac{(ax+b)^{m/2}}{x} dx = \frac{2(ax+b)^{m/2}}{m} + b \int \frac{(ax+b)^{(m-2)/2}}{x} dx$$

$$1.2.20 \quad \int \frac{(ax+b)^{m/2}}{x^2} dx = -\frac{(ax+b)^{(m+2)/2}}{bx} + \frac{ma}{2b} \int \frac{(ax+b)^{m/2}}{x} dx$$

$$1.2.21 \quad \int \frac{dx}{x(ax+b)^{m/2}} = \frac{2}{(m-2)b(ax+b)^{(m-2)/2}} + \frac{1}{b} \int \frac{dx}{x(ax+b)^{(m-2)/2}}$$

(3)

INTEGRALES QUE CONTIENEN $ax+b$ Y $px+q$

$$1.3.1 \quad \int \frac{dx}{(ax+b)(px+q)} = \frac{1}{bp-aq} \ln \left(\frac{px+q}{ax+b} \right)$$

$$1.3.2 \quad \int \frac{x dx}{(ax+b)(px+q)} = \frac{1}{bp-aq} \left\{ \frac{b}{a} \ln(ax+b) - \frac{q}{p} \ln(px+q) \right\}$$

$$1.3.3 \quad \int \frac{dx}{(ax+b)^2(px+q)} = \frac{1}{bp-aq} \left\{ \frac{1}{ax+b} + \frac{p}{bp-aq} \ln \left(\frac{px+q}{ax+b} \right) \right\}$$

$$1.3.4 \quad \int \frac{x dx}{(ax+b)^2(px+q)} = \frac{1}{bp-aq} \left\{ \frac{q}{bp-aq} \ln \left(\frac{ax+b}{px+q} \right) - \frac{b}{a(ax+b)} \right\}$$

$$1.3.5 \quad \int \frac{x^2 dx}{(ax+b)^2(px+q)} = \frac{b^2}{(bp-aq)a^2(ax+b)} + \frac{1}{(bp-aq)^2} \left\{ \frac{q^2}{p} \ln(px+q) + \frac{b(bp-2aq)}{a^2} \ln(ax+b) \right\}$$

$$1.3.6 \quad \int \frac{dx}{(ax+b)^m(px+q)^n} = \frac{-1}{(n-1)(bp-aq)} \left\{ \frac{1}{(ax+b)^{m-1}(px+q)^{n-1}} \right. \\ \left. + a(m+n-2) \int \frac{dx}{(ax+b)^m(px+q)^{n-1}} \right\}$$

$$1.3.7 \quad \int \frac{ax+b}{px+q} dx = \frac{ax}{p} + \frac{bp-aq}{p^2} \ln(px+q)$$

$$1.3.8 \quad \int \frac{(ax+b)^m}{(px+q)^n} dx = \begin{cases} \frac{-1}{(n-1)(bp-aq)} \left\{ \frac{(ax+b)^{m+1}}{(px+q)^{n-1}} + (n-m-2)a \int \frac{(ax+b)^m}{(px+q)^{n-1}} dx \right\} \\ \frac{-1}{(n-m-1)p} \left\{ \frac{(ax+b)^m}{(px+q)^{n-1}} + m(bp-aq) \int \frac{(ax+b)^{m-1}}{(px+q)^n} dx \right\} \\ \frac{-1}{(n-1)p} \left\{ \frac{(ax+b)^m}{(px+q)^{n-1}} - ma \int \frac{(ax+b)^{m-1}}{(px+q)^{n-1}} dx \right\} \end{cases}$$

(4) INTEGRALES QUE CONTIENEN $\sqrt{ax+b}$ Y $px+q$

$$1.4.1 \quad \int \frac{px+q}{\sqrt{ax+b}} dx = \frac{2(apx+3aq-2bp)}{3a^2} \sqrt{ax+b}$$

$$1.4.2 \quad \int \frac{dx}{(px+q)\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{bp-aq}\sqrt{p}} \ln \left(\frac{\sqrt{p(ax+b)} - \sqrt{bp-aq}}{\sqrt{p(ax+b)} + \sqrt{bp-aq}} \right) \\ \frac{2}{\sqrt{aq-bp}\sqrt{p}} \operatorname{tg}^{-1} \sqrt{\frac{p(ax+b)}{aq-bp}} \end{cases}$$

$$1.4.3 \quad \int \frac{\sqrt{ax+b}}{px+q} dx = \begin{cases} \frac{2\sqrt{ax+b}}{p} + \frac{\sqrt{bp-aq}}{p\sqrt{p}} \ln \left(\frac{\sqrt{p(ax+b)} - \sqrt{bp-aq}}{\sqrt{p(ax+b)} + \sqrt{bp-aq}} \right) \\ \frac{2\sqrt{ax+b}}{2} - \frac{2\sqrt{aq-bp}}{p\sqrt{p}} \operatorname{tg}^{-1} \sqrt{\frac{p(ax+b)}{aq-bp}} \end{cases}$$

$$1.4.4 \quad \int (px+q)^n \sqrt{ax+b} dx = \frac{2(px+q)^{n+1} \sqrt{ax+b}}{(2n+3)p} + \frac{bp-aq}{(2n+3)p} \int \frac{(px+q)^n}{\sqrt{ax+b}} dx$$

$$1.4.5 \quad \int \frac{dx}{(px+q)^n \sqrt{ax+b}} = \frac{\sqrt{ax+b}}{(n-1)(aq-bp)(px+q)^{n-1}} + \frac{(2n-3)a}{2(n-1)(aq-bp)} \int \frac{dx}{(px+q)^{n-1} \sqrt{ax+b}}$$

$$1.4.6 \quad \int \frac{(px+q)^n}{\sqrt{ax+b}} dx = \frac{2(px+q)^n \sqrt{ax+b}}{(2n+1)a} + \frac{2n(aq-bp)}{(2n+1)a} \int \frac{(px+q)^{n-1} dx}{\sqrt{ax+b}}$$

$$1.4.7 \quad \int \frac{\sqrt{ax+b}}{(px+q)^n} dx = \frac{-\sqrt{ax+b}}{(n-1)p(px+q)^{n-1}} + \frac{a}{2(n-1)p} \int \frac{dx}{(px+q)^{n-1} \sqrt{ax+b}}$$

(5) INTEGRALES QUE CONTIENEN $\sqrt{ax+b}$ Y $\sqrt{px+q}$

$$1.5.1 \quad \int \frac{dx}{\sqrt{(ax+b)(px+q)}} = \begin{cases} \frac{2}{\sqrt{ap}} \ln(\sqrt{a(px+q)} + \sqrt{p(ax+b)}) \\ \frac{2}{\sqrt{-ap}} \operatorname{tg}^{-1} \sqrt{\frac{-p(ax+b)}{a(px+q)}} \end{cases}$$

$$1.5.2 \quad \int \frac{x dx}{\sqrt{(ax+b)(px+q)}} = \frac{\sqrt{(ax+b)(px+q)}}{ap} - \frac{bp+aq}{2ap} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$1.5.3 \quad \int \sqrt{(ax+b)(px+q)} dx = \frac{2apx+bp+aq}{4ap} \sqrt{(ax+b)(px+q)} - \frac{(bp-aq)^2}{8ap} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$1.5.4 \quad \int \sqrt{\frac{px+q}{ax+b}} dx = \frac{\sqrt{(ax+b)(px+q)}}{a} + \frac{aq-bp}{2a} \int \frac{dx}{\sqrt{(ax+b)(px+q)}}$$

$$1.5.5 \quad \int \frac{dx}{(px+q)\sqrt{(ax+b)(px+q)}} = \frac{2\sqrt{ax+b}}{(aq-bp)\sqrt{px+q}}$$

(6)

INTEGRALES QUE CONTIENEN $x^2 + a^2$

$$1.6.1 \quad \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.2 \quad \int \frac{x dx}{x^2 + a^2} = \frac{1}{2} \ln(x^2 + a^2)$$

$$1.6.3 \quad \int \frac{x^2 dx}{x^2 + a^2} = x - a \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.4 \quad \int \frac{x^3 dx}{x^2 + a^2} = \frac{x^2}{2} - \frac{a^2}{2} \ln(x^2 + a^2)$$

$$1.6.5 \quad \int \frac{dx}{x(x^2 + a^2)} = \frac{1}{2a^2} \ln\left(\frac{x^2}{x^2 + a^2}\right)$$

$$1.6.6 \quad \int \frac{dx}{x^2(x^2 + a^2)} = -\frac{1}{a^2 x} - \frac{1}{a^3} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.7 \quad \int \frac{dx}{x^3(x^2 + a^2)} = -\frac{1}{2a^2 x^2} - \frac{1}{2a^4} \ln\left(\frac{x^2}{x^2 + a^2}\right)$$

$$1.6.8 \quad \int \frac{dx}{(x^2 + a^2)^2} = \frac{x}{2a^2(x^2 + a^2)} + \frac{1}{2a^3} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.9 \quad \int \frac{x dx}{(x^2 + a^2)^2} = \frac{-1}{2(x^2 + a^2)}$$

$$1.6.10 \quad \int \frac{x^2 dx}{(x^2 + a^2)^2} = \frac{-x}{2(x^2 + a^2)} + \frac{1}{2a} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.11 \quad \int \frac{x^3 dx}{(x^2 + a^2)^2} = \frac{a^2}{2(x^2 + a^2)} + \frac{1}{2} \ln(x^2 + a^2)$$

$$1.6.12 \quad \int \frac{dx}{x(x^2 + a^2)^2} = \frac{1}{2a^2(x^2 + a^2)} + \frac{1}{2a^4} \ln\left(\frac{x^2}{x^2 + a^2}\right)$$

$$1.6.13 \quad \int \frac{dx}{x^2(x^2 + a^2)^2} = -\frac{1}{a^4 x} - \frac{x}{2a^4(x^2 + a^2)} - \frac{3}{2a^5} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.6.14 \quad \int \frac{dx}{x^3(x^2 + a^2)^2} = -\frac{1}{2a^4x^2} - \frac{1}{2a^4(x^2 + a^2)} - \frac{1}{a^6} \ln\left(\frac{x^2}{x^2 + a^2}\right)$$

$$1.6.15 \quad \int \frac{dx}{(x^2 + a^2)^n} = \frac{x}{2(n-1)a^2(x^2 + a^2)^{n-1}} + \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(x^2 + a^2)^{n-1}}$$

$$1.6.16 \quad \int \frac{x dx}{(x^2 + a^2)^n} = \frac{-1}{2(n-1)(x^2 + a^2)^{n-1}}$$

$$1.6.17 \quad \int \frac{dx}{x(x^2 + a^2)^n} = \frac{1}{2(n-1)a^2(x^2 + a^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x(x^2 + a^2)^{n-1}}$$

$$1.6.18 \quad \int \frac{x^m dx}{(x^2 + a^2)^n} = \int \frac{x^{m-2} dx}{(x^2 + a^2)^{n-1}} - a^2 \int \frac{x^{m-2} dx}{(x^2 + a^2)^n}$$

$$1.6.19 \quad \int \frac{dx}{x^m(x^2 + a^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^m(x^2 + a^2)^{n-1}} - \frac{1}{a^2} \int \frac{dx}{x^{m-2}(x^2 + a^2)^n}$$

(7) INTEGRALES QUE CONTIENEN $x^2 - a^2, x^2 > a^2$

$$1.7.1 \quad \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right) \quad \text{o} \quad -\frac{1}{a} \coth^{-1} \frac{x}{a}$$

$$1.7.2 \quad \int \frac{x dx}{x^2 - a^2} = \frac{1}{2} \ln(x^2 - a^2)$$

$$1.7.3 \quad \int \frac{x^2 dx}{x^2 - a^2} = x + \frac{a}{2} \ln\left(\frac{x-a}{x+a}\right)$$

$$1.7.4 \quad \int \frac{x^3 dx}{x^2 - a^2} = \frac{x^2}{2} + \frac{a^2}{2} \ln(x^2 - a^2)$$

$$1.7.5 \quad \int \frac{dx}{x(x^2 - a^2)} = \frac{1}{2a^2} \ln\left(\frac{x^2 - a^2}{x^2}\right)$$

$$1.7.6 \quad \int \frac{dx}{x^2(x^2 - a^2)} = \frac{1}{a^2x} + \frac{1}{2a^3} \ln\left(\frac{x-a}{x+a}\right)$$

$$1.7.7 \quad \int \frac{dx}{x^3(x^2 - a^2)} = \frac{1}{2a^2x^2} - \frac{1}{2a^4} \ln\left(\frac{x^2}{x^2 - a^2}\right)$$

$$1.7.8 \quad \int \frac{dx}{(x^2 - a^2)^2} = \frac{-x}{2a^2(x^2 - a^2)} - \frac{1}{4a^3} \ln\left(\frac{x-a}{x+a}\right)$$

$$1.7.9 \quad \int \frac{x dx}{(x^2 - a^2)^2} = \frac{-1}{2(x^2 - a^2)}$$

$$1.7.10 \quad \int \frac{x^2 dx}{(x^2 - a^2)^2} = \frac{-x}{2(x^2 - a^2)} + \frac{1}{4a} \ln\left(\frac{x-a}{x+a}\right)$$

$$1.7.11 \quad \int \frac{x^3 dx}{(x^2 - a^2)^2} = \frac{-a}{2(x^2 - a^2)} + \frac{1}{2} \ln(x^2 - a^2)$$

$$1.7.12 \quad \int \frac{dx}{x(x^2 - a^2)^2} = \frac{-1}{2a^2(x^2 - a^2)} + \frac{1}{2a^4} \ln\left(\frac{x^2}{x^2 - a^2}\right)$$

$$1.7.13 \quad \int \frac{dx}{x^2(x^2 - a^2)^2} = -\frac{1}{a^4 x} - \frac{x}{2a^4(x^2 - a^2)} - \frac{3}{4a^5} \ln\left(\frac{x - a}{x + a}\right)$$

$$1.7.14 \quad \int \frac{dx}{x^3(x^2 - a^2)^2} = -\frac{1}{2a^4 x^2} - \frac{1}{2a^4(x^2 - a^2)} + \frac{1}{a^6} \ln\left(\frac{x^2}{x^2 - a^2}\right)$$

$$1.7.15 \quad \int \frac{dx}{(x^2 - a^2)^n} = \frac{-x}{2(n-1)a^2(x^2 - a^2)^{n-1}} - \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(x^2 - a^2)^{n-1}}$$

$$1.7.16 \quad \int \frac{x dx}{(x^2 - a^2)^n} = \frac{-1}{2(n-1)(x^2 - a^2)^{n-1}}$$

$$1.7.17 \quad \int \frac{dx}{x(x^2 - a^2)^n} = \frac{-1}{2(n-1)a^2(x^2 - a^2)^{n-1}} - \frac{1}{a^2} \int \frac{dx}{x(x^2 - a^2)^{n-1}}$$

$$1.7.18 \quad \int \frac{x^m dx}{(x^2 - a^2)^n} = \int \frac{x^{m-2} dx}{(x^2 - a^2)^{n-1}} + a^2 \int \frac{x^{m-2} dx}{(x^2 - a^2)^n}$$

$$1.7.19 \quad \int \frac{dx}{x^m(x^2 - a^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^{m-2}(x^2 - a^2)^n} - \frac{1}{a^2} \int \frac{dx}{x^m(x^2 - a^2)^{n-1}}$$

(8)

INTEGRALES QUE CONTIENEN $a^2 - x^2$, $x^2 < a^2$

$$1.8.1 \quad \int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right) \quad \text{o} \quad \frac{1}{a} \operatorname{th}^{-1} \frac{x}{a}$$

$$1.8.2 \quad \int \frac{x dx}{a^2 - x^2} = -\frac{1}{2} \ln(a^2 - x^2)$$

$$1.8.3 \quad \int \frac{x^2 dx}{a^2 - x^2} = -x + \frac{a}{2} \ln\left(\frac{a+x}{a-x}\right)$$

$$1.8.4 \quad \int \frac{x^3 dx}{a^2 - x^2} = -\frac{x^2}{2} - \frac{a^2}{2} \ln(a^2 - x^2)$$

$$1.8.5 \quad \int \frac{dx}{x(a^2 - x^2)} = \frac{1}{2a^2} \ln\left(\frac{x^2}{a^2 - x^2}\right)$$

$$1.8.6 \quad \int \frac{dx}{x^2(a^2 - x^2)} = -\frac{1}{a^2 x} + \frac{1}{2a^3} \ln\left(\frac{a+x}{a-x}\right)$$

$$1.8.7 \quad \int \frac{dx}{x^3(a^2 - x^2)} = -\frac{1}{2a^2 x^2} + \frac{1}{2a^4} \ln\left(\frac{x^2}{a^2 - x^2}\right)$$

$$1.8.8 \quad \int \frac{dx}{(a^2 - x^2)^2} = \frac{x}{2a^2(a^2 - x^2)} + \frac{1}{4a^3} \ln \left(\frac{a+x}{a-x} \right)$$

$$1.8.9 \quad \int \frac{x dx}{(a^2 - x^2)^2} = \frac{1}{2(a^2 - x^2)}$$

$$1.8.10 \quad \int \frac{x^2 dx}{(a^2 - x^2)^2} = \frac{x}{2(a^2 - x^2)} - \frac{1}{4a} \ln \left(\frac{a+x}{a-x} \right)$$

$$1.8.11 \quad \int \frac{x^3 dx}{(a^2 - x^2)^2} = \frac{a^2}{2(a^2 - x^2)} + \frac{1}{2} \ln(a^2 - x^2)$$

$$1.8.12 \quad \int \frac{dx}{x(a^2 - x^2)^2} = \frac{1}{2a^2(a^2 - x^2)} + \frac{1}{2a^4} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$1.8.13 \quad \int \frac{dx}{x^2(a^2 - x^2)^2} = \frac{-1}{a^4 x} + \frac{x}{2a^4(a^2 - x^2)} + \frac{3}{4a^5} \ln \left(\frac{a+x}{a-x} \right)$$

$$1.8.14 \quad \int \frac{dx}{x^3(a^2 - x^2)^2} = \frac{-1}{2a^4 x^2} + \frac{1}{2a^4(a^2 - x^2)} + \frac{1}{a^6} \ln \left(\frac{x^2}{a^2 - x^2} \right)$$

$$1.8.15 \quad \int \frac{dx}{(a^2 - x^2)^n} = \frac{x}{2(n-1)a^2(a^2 - x^2)^{n-1}} + \frac{2n-3}{(2n-2)a^2} \int \frac{dx}{(a^2 - x^2)^{n-1}}$$

$$1.8.16 \quad \int \frac{x dx}{(a^2 - x^2)^n} = \frac{1}{2(n-1)(a^2 - x^2)^{n-1}}$$

$$1.8.17 \quad \int \frac{dx}{x(a^2 - x^2)^n} = \frac{1}{2(n-1)a^2(a^2 - x^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x^2(a^2 - x^2)^{n-1}}$$

$$1.8.18 \quad \int \frac{x^m dx}{(a^2 - x^2)^n} = a^2 \int \frac{x^{m-2} dx}{(a^2 - x^2)^n} - \int \frac{x^{m-2} dx}{(a^2 - x^2)^{n-1}}$$

$$1.8.19 \quad \int \frac{dx}{x^m(a^2 - x^2)^n} = \frac{1}{a^2} \int \frac{dx}{x^m(a^2 - x^2)^{n-1}} + \frac{1}{a^2} \int \frac{dx}{x^{m-2}(a^2 - x^2)^n}$$

(9) INTEGRALES QUE CONTIENEN $\sqrt{x^2 + a^2}$

$$1.9.1 \quad \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{x^2 + a^2}) \quad \text{o} \quad \text{sh}^{-1} \frac{x}{a}$$

$$1.9.2 \quad \int \frac{x dx}{\sqrt{x^2 + a^2}} = \sqrt{x^2 + a^2}$$

$$1.9.3 \quad \int \frac{x^2 dx}{\sqrt{x^2 + a^2}} = \frac{x\sqrt{x^2 + a^2}}{2} - \frac{a^2}{2} \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.4 \quad \int \frac{x^3 dx}{\sqrt{x^2 + a^2}} = \frac{(x^2 + a^2)^{3/2}}{3} - a^2 \sqrt{x^2 + a^2}$$

$$1.9.5 \quad \int \frac{dx}{x \sqrt{x^2 + a^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.6 \quad \int \frac{dx}{x^2 \sqrt{x^2 + a^2}} = -\frac{\sqrt{x^2 + a^2}}{a^2 x}$$

$$1.9.7 \quad \int \frac{dx}{x^3 \sqrt{x^2 + a^2}} = -\frac{\sqrt{x^2 + a^2}}{2a^2 x^2} + \frac{1}{2a^3} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.8 \quad \int \sqrt{x^2 + a^2} dx = \frac{x \sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.9 \quad \int x \sqrt{x^2 + a^2} dx = \frac{(x^2 + a^2)^{3/2}}{3}$$

$$1.9.10 \quad \int x^2 \sqrt{x^2 + a^2} dx = \frac{x(x^2 + a^2)^{3/2}}{4} - \frac{a^2 x \sqrt{x^2 + a^2}}{8} - \frac{a^4}{8} \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.11 \quad \int x^3 \sqrt{x^2 + a^2} dx = \frac{(x^2 + a^2)^{5/2}}{5} - \frac{a^2 (x^2 + a^2)^{3/2}}{3}$$

$$1.9.12 \quad \int \frac{\sqrt{x^2 + a^2}}{x} dx = \sqrt{x^2 + a^2} - a \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.13 \quad \int \frac{\sqrt{x^2 + a^2}}{x^2} dx = -\frac{\sqrt{x^2 + a^2}}{x} + \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.14 \quad \int \frac{\sqrt{x^2 + a^2}}{x^3} dx = -\frac{\sqrt{x^2 + a^2}}{2x^2} - \frac{1}{2a} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.15 \quad \int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2 + a^2}}$$

$$1.9.16 \quad \int \frac{x dx}{(x^2 + a^2)^{3/2}} = \frac{-1}{\sqrt{x^2 + a^2}}$$

$$1.9.17 \quad \int \frac{x^2 dx}{(x^2 + a^2)^{3/2}} = \frac{-x}{\sqrt{x^2 + a^2}} + \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.18 \quad \int \frac{x^3 dx}{(x^2 + a^2)^{3/2}} = \sqrt{x^2 + a^2} + \frac{a^2}{\sqrt{x^2 + a^2}}$$

$$1.9.19 \quad \int \frac{dx}{x(x^2 + a^2)^{3/2}} = \frac{1}{a^2 \sqrt{x^2 + a^2}} - \frac{1}{a^3} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.20 \quad \int \frac{dx}{x^2(x^2 + a^2)^{3/2}} = -\frac{\sqrt{x^2 + a^2}}{a^4 x} - \frac{x}{a^4 \sqrt{x^2 + a^2}}$$

$$1.9.21 \quad \int \frac{dx}{x^3(x^2 + a^2)^{3/2}} = \frac{-1}{2a^2 x^2 \sqrt{x^2 + a^2}} - \frac{3}{2a^4 \sqrt{x^2 + a^2}} + \frac{3}{2a^5} \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.22 \quad \int (x^2 + a^2)^{3/2} dx = \frac{x(x^2 + a^2)^{3/2}}{4} + \frac{3a^2 x \sqrt{x^2 + a^2}}{8} + \frac{3}{8} a^4 \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.23 \quad \int x(x^2 + a^2)^{3/2} dx = \frac{(x^2 + a^2)^{5/2}}{5}$$

$$1.9.24 \quad \int x^2(x^2 + a^2)^{3/2} dx = \frac{x(x^2 + a^2)^{5/2}}{6} - \frac{a^2 x(x^2 + a^2)^{3/2}}{24} - \frac{a^4 x \sqrt{x^2 + a^2}}{16} - \frac{a^6}{16} \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.25 \quad \int x^3(x^2 + a^2)^{3/2} dx = \frac{(x^2 + a^2)^{7/2}}{7} - \frac{a^2(x^2 + a^2)^{5/2}}{5}$$

$$1.9.26 \quad \int \frac{(x^2 + a^2)^{3/2}}{x} dx = \frac{(x^2 + a^2)^{3/2}}{3} + a^2 \sqrt{x^2 + a^2} - a^3 \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

$$1.9.27 \quad \int \frac{(x^2 + a^2)^{3/2}}{x^2} dx = -\frac{(x^2 + a^2)^{3/2}}{x} + \frac{3x \sqrt{x^2 + a^2}}{2} + \frac{3}{2} a^2 \ln(x + \sqrt{x^2 + a^2})$$

$$1.9.28 \quad \int \frac{(x^2 + a^2)^{3/2}}{x^3} dx = -\frac{(x^2 + a^2)^{3/2}}{2x^2} + \frac{3}{2} \sqrt{x^2 + a^2} - \frac{3}{2} a \ln \left(\frac{a + \sqrt{x^2 + a^2}}{x} \right)$$

(10)

 INTEGRALES QUE CONTIENEN $\sqrt{x^2 - a^2}$

$$1.10.1 \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln(x + \sqrt{x^2 - a^2}), \quad \int \frac{x dx}{\sqrt{x^2 - a^2}} = \sqrt{x^2 - a^2}$$

$$1.10.2 \quad \int \frac{x^2 dx}{\sqrt{x^2 - a^2}} = \frac{x \sqrt{x^2 - a^2}}{2} + \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.3 \quad \int \frac{x^3 dx}{\sqrt{x^2 - a^2}} = \frac{(x^2 - a^2)^{3/2}}{3} + a^2 \sqrt{x^2 - a^2}$$

$$1.10.4 \quad \int \frac{dx}{x \sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.5 \quad \int \frac{dx}{x^2 \sqrt{x^2 - a^2}} = \frac{\sqrt{x^2 - a^2}}{a^2 x}$$

$$1.10.6 \quad \int \frac{dx}{x^3 \sqrt{x^2 - a^2}} = \frac{\sqrt{x^2 - a^2}}{2a^2 x^2} + \frac{1}{2a^3} \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.7 \quad \int \sqrt{x^2 - a^2} dx = \frac{x \sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.8 \quad \int x \sqrt{x^2 - a^2} dx = \frac{(x^2 - a^2)^{3/2}}{3}$$

$$1.10.9 \quad \int x^2 \sqrt{x^2 - a^2} dx = \frac{x(x^2 - a^2)^{3/2}}{4} + \frac{a^2 x \sqrt{x^2 - a^2}}{8} - \frac{a^4}{8} \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.10 \quad \int x^3 \sqrt{x^2 - a^2} dx = \frac{(x^2 - a^2)^{5/2}}{5} + \frac{a^2(x^2 - a^2)^{3/2}}{3}$$

$$1.10.11 \quad \int \frac{\sqrt{x^2 - a^2}}{x} dx = \sqrt{x^2 - a^2} - a \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.12 \quad \int \frac{\sqrt{x^2 - a^2}}{x^2} dx = -\frac{\sqrt{x^2 - a^2}}{x} + \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.13 \quad \int \frac{\sqrt{x^2 - a^2}}{x^3} dx = -\frac{\sqrt{x^2 - a^2}}{2x^2} + \frac{1}{2a} \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.14 \quad \int \frac{dx}{(x^2 - a^2)^{3/2}} = -\frac{x}{a^2 \sqrt{x^2 - a^2}}$$

$$1.10.15 \quad \int \frac{x dx}{(x^2 - a^2)^{3/2}} = \frac{-1}{\sqrt{x^2 - a^2}}$$

$$1.10.16 \quad \int \frac{x^2 dx}{(x^2 - a^2)^{3/2}} = -\frac{x}{\sqrt{x^2 - a^2}} + \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.17 \quad \int \frac{x^3 dx}{(x^2 - a^2)^{3/2}} = \sqrt{x^2 - a^2} - \frac{a^2}{\sqrt{x^2 - a^2}}$$

$$1.10.18 \quad \int \frac{dx}{x(x^2 - a^2)^{3/2}} = \frac{-1}{a^2 \sqrt{x^2 - a^2}} - \frac{1}{a^3} \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.19 \quad \int \frac{dx}{x^2(x^2 - a^2)^{3/2}} = -\frac{\sqrt{x^2 - a^2}}{a^4 x} - \frac{x}{a^4 \sqrt{x^2 - a^2}}$$

$$1.10.20 \quad \int \frac{dx}{x^3(x^2 - a^2)^{3/2}} = \frac{1}{2a^2 x^2 \sqrt{x^2 - a^2}} - \frac{3}{2a^4 \sqrt{x^2 - a^2}} - \frac{3}{2a^5} \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.21 \quad \int (x^2 - a^2)^{3/2} dx = \frac{x(x^2 - a^2)^{3/2}}{4} - \frac{3a^2 x \sqrt{x^2 - a^2}}{8} + \frac{3}{8} a^4 \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.22 \quad \int x(x^2 - a^2)^{3/2} dx = \frac{(x^2 - a^2)^{5/2}}{5}$$

$$1.10.23 \quad \int x^2(x^2 - a^2)^{3/2} dx = \frac{x(x^2 - a^2)^{5/2}}{6} + \frac{a^2 x(x^2 - a^2)^{3/2}}{24} - \frac{a^4 x \sqrt{x^2 - a^2}}{16} + \frac{a^6}{16} \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.24 \quad \int x^3(x^2 - a^2)^{3/2} dx = \frac{(x^2 - a^2)^{7/2}}{7} + \frac{a^2(x^2 - a^2)^{5/2}}{5}$$

$$1.10.25 \quad \int \frac{(x^2 - a^2)^{3/2}}{x} dx = \frac{(x^2 - a^2)^{3/2}}{3} - a^2 \sqrt{x^2 - a^2} + a^3 \sec^{-1} \left| \frac{x}{a} \right|$$

$$1.10.26 \quad \int \frac{(x^2 - a^2)^{3/2}}{x^2} dx = -\frac{(x^2 - a^2)^{3/2}}{x} + \frac{3x \sqrt{x^2 - a^2}}{2} - \frac{3}{2} a^2 \ln(x + \sqrt{x^2 - a^2})$$

$$1.10.27 \quad \int \frac{(x^2 - a^2)^{3/2}}{x^3} dx = -\frac{(x^2 - a^2)^{3/2}}{2x^2} + \frac{3 \sqrt{x^2 - a^2}}{2} - \frac{3}{2} a \sec^{-1} \left| \frac{x}{a} \right|$$

(11) INTEGRALES QUE CONTIENEN $\sqrt{a^2 - x^2}$

$$1.11.1 \quad \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$$

$$1.11.2 \quad \int \frac{x dx}{\sqrt{a^2 - x^2}} = -\sqrt{a^2 - x^2}$$

$$1.11.3 \quad \int \frac{x^2 dx}{\sqrt{a^2 - x^2}} = -\frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$1.11.4 \quad \int \frac{x^3 dx}{\sqrt{a^2 - x^2}} = \frac{(a^2 - x^2)^{3/2}}{3} - a^2 \sqrt{a^2 - x^2}$$

$$1.11.5 \quad \int \frac{dx}{x \sqrt{a^2 - x^2}} = -\frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.6 \quad \int \frac{dx}{x^2 \sqrt{a^2 - x^2}} = -\frac{\sqrt{a^2 - x^2}}{a^2 x}$$

$$1.11.7 \quad \int \frac{dx}{x^3 \sqrt{a^2 - x^2}} = -\frac{\sqrt{a^2 - x^2}}{2a^2 x^2} - \frac{1}{2a^3} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.8 \quad \int \sqrt{a^2 - x^2} dx = \frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$1.11.9 \quad \int x \sqrt{a^2 - x^2} dx = -\frac{(a^2 - x^2)^{3/2}}{3}$$

$$1.11.10 \quad \int x^2 \sqrt{a^2 - x^2} dx = -\frac{x(a^2 - x^2)^{3/2}}{4} + \frac{a^2 x \sqrt{a^2 - x^2}}{8} + \frac{a^4}{8} \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.11 \quad \int x^3 \sqrt{a^2 - x^2} dx = \frac{(a^2 - x^2)^{5/2}}{5} - \frac{a^2(a^2 - x^2)^{3/2}}{3}$$

$$1.11.12 \quad \int \frac{\sqrt{a^2 - x^2}}{x} dx = \sqrt{a^2 - x^2} - a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.13 \quad \int \frac{\sqrt{a^2 - x^2}}{x^2} dx = -\frac{\sqrt{a^2 - x^2}}{x} - \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.14 \quad \int \frac{\sqrt{a^2 - x^2}}{x^3} dx = -\frac{\sqrt{a^2 - x^2}}{2x^2} + \frac{1}{2a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.15 \quad \int \frac{dx}{(a^2 - x^2)^{3/2}} = \frac{x}{a^2 \sqrt{a^2 - x^2}}$$

$$1.11.16 \quad \int \frac{x dx}{(a^2 - x^2)^{3/2}} = \frac{1}{\sqrt{a^2 - x^2}}$$

$$1.11.17 \quad \int \frac{x^2 dx}{(a^2 - x^2)^{3/2}} = \frac{x}{\sqrt{a^2 - x^2}} - \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.18 \quad \int \frac{x^3 dx}{(a^2 - x^2)^{3/2}} = \sqrt{a^2 - x^2} + \frac{a^2}{\sqrt{a^2 - x^2}}$$

$$1.11.19 \quad \int \frac{dx}{x(a^2 - x^2)^{3/2}} = \frac{1}{a^2 \sqrt{a^2 - x^2}} - \frac{1}{a^3} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.20 \quad \int \frac{dx}{x^2(a^2 - x^2)^{3/2}} = -\frac{\sqrt{a^2 - x^2}}{a^4 x} + \frac{x}{a^4 \sqrt{a^2 - x^2}}$$

$$1.11.21 \quad \int \frac{dx}{x^3(a^2 - x^2)^{3/2}} = \frac{-1}{2a^2 x^2 \sqrt{a^2 - x^2}} + \frac{3}{2a^4 \sqrt{a^2 - x^2}} - \frac{3}{2a^5} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.22 \quad \int (a^2 - x^2)^{3/2} dx = \frac{x(a^2 - x^2)^{3/2}}{4} + \frac{3a^2 x \sqrt{a^2 - x^2}}{8} + \frac{3}{8} a^4 \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.23 \quad \int x(a^2 - x^2)^{3/2} dx = -\frac{(a^2 - x^2)^{5/2}}{5}$$

$$1.11.24 \quad \int x^2(a^2 - x^2)^{3/2} dx = -\frac{x(a^2 - x^2)^{5/2}}{6} + \frac{a^2 x(a^2 - x^2)^{3/2}}{24} + \frac{a^4 x \sqrt{a^2 - x^2}}{16} + \frac{a^6}{16} \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.25 \quad \int x^3(a^2 - x^2)^{3/2} dx = \frac{(a^2 - x^2)^{7/2}}{7} - \frac{a^2(a^2 - x^2)^{5/2}}{5}$$

$$1.11.26 \quad \int \frac{(a^2 - x^2)^{3/2}}{x} dx = \frac{(a^2 - x^2)^{3/2}}{3} + a^2 \sqrt{a^2 - x^2} - a^3 \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.11.27 \quad \int \frac{(a^2 - x^2)^{3/2}}{x^2} dx = -\frac{(a^2 - x^2)^{3/2}}{x} - \frac{3x\sqrt{a^2 - x^2}}{2} + \frac{3}{2} a^2 \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.11.28 \quad \int \frac{(a^2 - x^2)^{3/2}}{x^3} dx = -\frac{(a^2 - x^2)^{3/2}}{2x^2} - \frac{3\sqrt{a^2 - x^2}}{2} + \frac{3}{2} a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

(12) INTEGRALES QUE CONTIENEN $ax^2 + bx + c$

$$1.12.1 \quad \int \frac{dx}{ax^2 + bx + c} = \begin{cases} \frac{2}{\sqrt{4ac - b^2}} \operatorname{tg}^{-1} \frac{2ax + b}{\sqrt{4ac - b^2}} \\ \frac{1}{\sqrt{b^2 - 4ac}} \ln \left(\frac{2ax + b - \sqrt{b^2 - 4ac}}{2ax + b + \sqrt{b^2 - 4ac}} \right) \end{cases}$$

Si $b^2 = 4ac$, entonces $ax^2 + bx + c = a(x + b/2a)^2$ y se pueden utilizar los resultados de las integrales 1.1.6 a 1.1.10 y 1.1.14 a 1.1.17. Si $b = 0$, utilícense los resultados de las páginas 253-254. Si $a = 0$ o si $c = 0$, pueden usarse los resultados de las páginas 249-250.

$$1.12.2 \quad \int \frac{x dx}{ax^2 + bx + c} = \frac{1}{2a} \ln(ax^2 + bx + c) - \frac{b}{2a} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.3 \quad \int \frac{x^2 dx}{ax^2 + bx + c} = \frac{x}{a} - \frac{b}{2a^2} \ln(ax^2 + bx + c) + \frac{b^2 - 2ac}{2a^2} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.4 \quad \int \frac{x^m dx}{ax^2 + bx + c} = \frac{x^{m-1}}{(m-1)a} - \frac{c}{a} \int \frac{x^{m-2} dx}{ax^2 + bx + c} - \frac{b}{a} \int \frac{x^{m-1} dx}{ax^2 + bx + c}$$

$$1.12.5 \quad \int \frac{dx}{x(ax^2 + bx + c)} = \frac{1}{2c} \ln \left(\frac{x^2}{ax^2 + bx + c} \right) - \frac{b}{2c} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.6 \quad \int \frac{dx}{x^2(ax^2 + bx + c)} = \frac{b}{2c^2} \ln \left(\frac{ax^2 + bx + c}{x^2} \right) - \frac{1}{cx} + \frac{b^2 - 2ac}{2c^2} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.7 \quad \int \frac{dx}{x^n(ax^2 + bx + c)} = -\frac{1}{(n-1)cx^{n-1}} - \frac{b}{c} \int \frac{dx}{x^{n-1}(ax^2 + bx + c)} - \frac{a}{c} \int \frac{dx}{x^{n-2}(ax^2 + bx + c)}$$

$$1.12.8 \quad \int \frac{dx}{(ax^2 + bx + c)^2} = \frac{2ax + b}{(4ac - b^2)(ax^2 + bx + c)} + \frac{2a}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.9 \quad \int \frac{x dx}{(ax^2 + bx + c)^2} = -\frac{bx + 2c}{(4ac - b^2)(ax^2 + bx + c)} - \frac{b}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.10 \quad \int \frac{x^2 dx}{(ax^2 + bx + c)^2} = \frac{(b^2 - 2ac)x + bc}{a(4ac + b^2)(ax^2 + bx + c)} + \frac{2c}{4ac - b^2} \int \frac{dx}{ax^2 + bx + c}$$

$$1.12.11 \quad \int \frac{x^m dx}{(ax^2 + bx + c)^n} = -\frac{x^{m-1}}{(2n-m-1)a(ax^2 + bx + c)^{n-1}} + \frac{(m-1)c}{(2n-m-1)a} \int \frac{x^{m-2} dx}{(ax^2 + bx + c)^n} \\ - \frac{(n-m)b}{(2n-m-1)a} \int \frac{x^{m-1} dx}{(ax^2 + bx + c)^n}$$

$$1.12.12 \quad \int \frac{x^{2n-1} dx}{(ax^2 + bx + c)^n} = \frac{1}{a} \int \frac{x^{2n-3} dx}{(ax^2 + bx + c)^{n-1}} - \frac{c}{a} \int \frac{x^{2n-3} dx}{(ax^2 + bx + c)^n} - \frac{b}{a} \int \frac{x^{2n-2} dx}{(ax^2 + bx + c)^n}$$

$$1.12.13 \quad \int \frac{dx}{x(ax^2 + bx + c)^2} = \frac{1}{2c(ax^2 + bx + c)} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^2} + \frac{1}{c} \int \frac{dx}{x(ax^2 + bx + c)}$$

$$1.12.14 \quad \int \frac{dx}{x^2(ax^2 + bx + c)^2} = -\frac{1}{cx(ax^2 + bx + c)} - \frac{3a}{c} \int \frac{dx}{(ax^2 + bx + c)^2} - \frac{2b}{c} \int \frac{dx}{x(ax^2 + bx + c)^2}$$

$$1.12.15 \quad \int \frac{dx}{x^m(ax^2 + bx + c)^n} = -\frac{1}{(m-1)cx^{m-1}(ax^2 + bx + c)^{n-1}} - \frac{(m+2n-3)a}{(m-1)c} \int \frac{dx}{x^{m-2}(ax^2 + bx + c)^n} \\ - \frac{(m+n-2)b}{(m-1)c} \int \frac{dx}{x^{m-1}(ax^2 + bx + c)^n}$$

(13)

INTEGRALES QUE CONTIENEN $\sqrt{ax^2 + bx + c}$

En los resultados de este apartado, si $b^2 = 4ac$, es $\sqrt{ax^2 + bx + c} = \sqrt{a}(x + b/2a)$, y se pueden utilizar los resultados del Apartado 1 de esta tabla de integrales. Si $b = 0$, pueden aplicarse los resultados del Apartado 9 de esta tabla. Si $a = 0$ o $c = 0$, pueden usarse las integrales de los Apartados 2 y 5 de esta tabla.

$$1.13.1 \quad \int \frac{dx}{\sqrt{ax^2 + bx + c}} = \begin{cases} \frac{1}{\sqrt{a}} \ln(2\sqrt{a}\sqrt{ax^2 + bx + c} + 2ax + b) \\ -\frac{1}{\sqrt{-a}} \operatorname{sen}^{-1}\left(\frac{2ax + b}{\sqrt{b^2 - 4ac}}\right) \text{ o } \frac{1}{\sqrt{a}} \operatorname{sh}^{-1}\left(\frac{2ax + b}{\sqrt{4ac - b^2}}\right) \end{cases}$$

$$1.13.2 \quad \int \frac{x dx}{\sqrt{ax^2 + bx + c}} = \frac{\sqrt{ax^2 + bx + c}}{a} - \frac{b}{2a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$1.13.3 \quad \int \frac{x^2 dx}{\sqrt{ax^2 + bx + c}} = \frac{2ax - 3b}{4a^2} \sqrt{ax^2 + bx + c} + \frac{3b^2 - 4ac}{8a^2} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$1.13.4 \quad \int \frac{dx}{x\sqrt{ax^2 + bx + c}} = \begin{cases} -\frac{1}{\sqrt{c}} \ln\left(\frac{2\sqrt{c}\sqrt{ax^2 + bx + c} + bx + 2c}{x}\right) \\ \frac{1}{\sqrt{-c}} \operatorname{sen}^{-1}\left(\frac{bx + 2c}{|x|\sqrt{b^2 - 4ac}}\right) \text{ o } -\frac{1}{\sqrt{c}} \operatorname{sh}^{-1}\left(\frac{bx + 2c}{|x|\sqrt{4ac - b^2}}\right) \end{cases}$$

$$1.13.5 \quad \int \frac{dx}{x^2\sqrt{ax^2 + bx + c}} = -\frac{\sqrt{ax^2 + bx + c}}{cx} - \frac{b}{2c} \int \frac{dx}{x\sqrt{ax^2 + bx + c}}$$

$$1.13.6 \quad \int \sqrt{ax^2 + bx + c} dx = \frac{(2ax + b)\sqrt{ax^2 + bx + c}}{4a} + \frac{4ac - b^2}{8a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$1.13.7 \quad \int x \sqrt{ax^2 + bx + c} dx = \frac{(ax^2 + bx + c)^{3/2}}{3a} - \frac{b(2ax + b)}{8a^2} \sqrt{ax^2 + bx + c} - \frac{b(4ac - b^2)}{16a^2} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$1.13.8 \quad \int x^2 \sqrt{ax^2 + bx + c} dx = \frac{6ax - 5b}{24a^2} (ax^2 + bx + c)^{3/2} + \frac{5b^2 - 4ac}{16a^2} \int \sqrt{ax^2 + bx + c} dx$$

$$1.13.9 \quad \int \frac{\sqrt{ax^2 + bx + c}}{x} dx = \sqrt{ax^2 + bx + c} + \frac{b}{2} \int \frac{dx}{\sqrt{ax^2 + bx + c}} + c \int \frac{dx}{x \sqrt{ax^2 + bx + c}}$$

$$1.13.10 \quad \int \frac{\sqrt{ax^2 + bx + c}}{x^2} dx = -\frac{\sqrt{ax^2 + bx + c}}{x} + a \int \frac{dx}{\sqrt{ax^2 + bx + c}} + \frac{b}{2} \int \frac{dx}{x \sqrt{ax^2 + bx + c}}$$

$$1.13.11 \quad \int \frac{dx}{(ax^2 + bx + c)^{3/2}} = \frac{2(2ax + b)}{(4ac - b^2)\sqrt{ax^2 + bx + c}}$$

$$1.13.12 \quad \int \frac{x dx}{(ax^2 + bx + c)^{3/2}} = \frac{2(bx + 2c)}{(b^2 - 4ac)\sqrt{ax^2 + bx + c}}$$

$$1.13.13 \quad \int \frac{x^2 dx}{(ax^2 + bx + c)^{3/2}} = \frac{(2b^2 - 4ac)x + 2bc}{a(4ac - b^2)\sqrt{ax^2 + bx + c}} + \frac{1}{a} \int \frac{dx}{\sqrt{ax^2 + bx + c}}$$

$$1.13.14 \quad \int \frac{dx}{x(ax^2 + bx + c)^{3/2}} = \frac{1}{c\sqrt{ax^2 + bx + c}} + \frac{1}{c} \int \frac{dx}{x\sqrt{ax^2 + bx + c}} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^{3/2}}$$

$$1.13.15 \quad \int \frac{dx}{x^2(ax^2 + bx + c)^{3/2}} = -\frac{ax^2 + 2bx + c}{c^2 x \sqrt{ax^2 + bx + c}} + \frac{b^2 - 2ac}{2c^2} \int \frac{dx}{(ax^2 + bx + c)^{3/2}} - \frac{3b}{2c^2} \int \frac{dx}{x \sqrt{ax^2 + bx + c}}$$

$$1.13.16 \quad \int (ax^2 + bx + c)^{n+1/2} dx = \frac{(2ax + b)(ax^2 + bx + c)^{n+1/2}}{4a(n+1)} + \frac{(2n+1)(4ac - b^2)}{8a(n+1)} \int (ax^2 + bx + c)^{n-1/2} dx$$

$$1.13.17 \quad \int x(ax^2 + bx + c)^{n+1/2} dx = \frac{(ax^2 + bx + c)^{n+3/2}}{a(2n+3)} - \frac{b}{2a} \int (ax^2 + bx + c)^{n+1/2} dx$$

$$1.13.18 \quad \int \frac{dx}{(ax^2 + bx + c)^{n+1/2}} = \frac{2(2ax + b)}{(2n-1)(4ac - b^2)(ax^2 + bx + c)^{n-1/2}} + \frac{8a(n-1)}{(2n-1)(4ac - b^2)} \int \frac{dx}{(ax^2 + bx + c)^{n-1/2}}$$

$$1.13.19 \quad \int \frac{dx}{x(ax^2 + bx + c)^{n+1/2}} = \frac{1}{(2n-1)c(ax^2 + bx + c)^{n-1/2}} \\ + \frac{1}{c} \int \frac{dx}{x(ax^2 + bx + c)^{n-1/2}} - \frac{b}{2c} \int \frac{dx}{(ax^2 + bx + c)^{n+1/2}}$$

(14)**INTEGRALES QUE CONTIENEN $x^3 + a^3$**

Para fórmulas que contienen $x^3 - a^3$, reemplace a por $-a$.

$$1.14.1 \quad \int \frac{dx}{x^3 + a^3} = \frac{1}{6a^2} \ln \frac{(x+a)^2}{x^2 - ax + a^2} + \frac{1}{a^2 \sqrt{3}} \operatorname{tg}^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$1.14.2 \quad \int \frac{x dx}{x^3 + a^3} = \frac{1}{6a} \ln \frac{x^2 - ax + a^2}{(x+a)^2} + \frac{1}{a\sqrt{3}} \operatorname{tg}^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$1.14.3 \quad \int \frac{x^2 dx}{x^3 + a^3} = \frac{1}{3} \ln(x^3 + a^3)$$

$$1.14.4 \quad \int \frac{dx}{x(x^3 + a^3)} = \frac{1}{3a^3} \ln \left(\frac{x^3}{x^3 + a^3} \right)$$

$$1.14.5 \quad \int \frac{dx}{x^2(x^3 + a^3)} = -\frac{1}{a^3 x} - \frac{1}{6a^4} \ln \frac{x^2 - ax + a^2}{(x+a)^2} - \frac{1}{a^4 \sqrt{3}} \operatorname{tg}^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$1.14.6 \quad \int \frac{dx}{(x^3 + a^3)^2} = \frac{x}{3a^3(x^3 + a^3)} + \frac{1}{9a^5} \ln \frac{(x+a)^2}{x^2 - ax + a^2} + \frac{2}{3a^5 \sqrt{3}} \operatorname{tg}^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$1.14.7 \quad \int \frac{x dx}{(x^3 + a^3)^2} = \frac{x^2}{3a^3(x^3 + a^3)} + \frac{1}{18a^4} \ln \frac{x^2 - ax + a^2}{(x+a)^2} + \frac{1}{3a^4 \sqrt{3}} \operatorname{tg}^{-1} \frac{2x-a}{a\sqrt{3}}$$

$$1.14.8 \quad \int \frac{x^2 dx}{(x^3 + a^3)^2} = -\frac{1}{3(x^3 + a^3)}$$

$$1.14.9 \quad \int \frac{dx}{x(x^3 + a^3)^2} = \frac{1}{3a^3(x^3 + a^3)} + \frac{1}{3a^6} \ln \left(\frac{x^3}{x^3 + a^3} \right)$$

$$1.14.10 \quad \int \frac{dx}{x^2(x^3 + a^3)^2} = -\frac{1}{a^6 x} - \frac{x^2}{3a^6(x^3 + a^3)} - \frac{4}{3a^6} \int \frac{x dx}{x^3 + a^3} \quad [\text{Véase 14.2}]$$

$$1.14.11 \quad \int \frac{x^m dx}{x^3 + a^3} = \frac{x^{m-2}}{m-2} - a^3 \int \frac{x^{m-3} dx}{x^3 + a^3}$$

$$1.14.12 \quad \int \frac{dx}{x^n(x^3 + a^3)} = \frac{-1}{a^3(n-1)x^{n-1}} - \frac{1}{a^3} \int \frac{dx}{x^{n-3}(x^3 + a^3)}$$

(15) INTEGRALES QUE CONTIENEN $x^4 \pm a^4$

$$1.15.1 \quad \int \frac{dx}{x^4 + a^4} = \frac{1}{4a^3\sqrt{2}} \ln \left(\frac{x^2 + ax\sqrt{2} + a^2}{x^2 - ax\sqrt{2} + a^2} \right) - \frac{1}{2a^3\sqrt{2}} \left[\operatorname{tg}^{-1} \left(1 - \frac{x\sqrt{2}}{a} \right) - \operatorname{tg}^{-1} \left(1 + \frac{x\sqrt{2}}{a} \right) \right]$$

$$1.15.2 \quad \int \frac{x dx}{x^4 + a^4} = \frac{1}{2a^2} \operatorname{tg}^{-1} \frac{x^2}{a^2}$$

$$1.15.3 \quad \int \frac{x^2 dx}{x^4 + a^4} = \frac{1}{4a\sqrt{2}} \ln \left(\frac{x^2 - ax\sqrt{2} + a^2}{x^2 + ax\sqrt{2} + a^2} \right) - \frac{1}{2a\sqrt{2}} \left[\operatorname{tg}^{-1} \left(1 - \frac{x\sqrt{2}}{a} \right) - \operatorname{tg}^{-1} \left(1 + \frac{x\sqrt{2}}{a} \right) \right]$$

$$1.15.4 \quad \int \frac{x^3 dx}{x^4 + a^4} = \frac{1}{4} \ln(x^4 + a^4)$$

$$1.15.5 \quad \int \frac{dx}{x(x^4 + a^4)} = \frac{1}{4a^4} \ln \left(\frac{x^4}{x^4 + a^4} \right)$$

$$1.15.6 \quad \int \frac{dx}{x^2(x^4 + a^4)} = -\frac{1}{a^4 x} - \frac{1}{4a^5\sqrt{2}} \ln \left(\frac{x^2 - ax\sqrt{2} + a^2}{x^2 + ax\sqrt{2} + a^2} \right) + \frac{1}{2a^5\sqrt{2}} \left[\operatorname{tg}^{-1} \left(1 - \frac{x\sqrt{2}}{a} \right) - \operatorname{tg}^{-1} \left(1 + \frac{x\sqrt{2}}{a} \right) \right]$$

$$1.15.7 \quad \int \frac{dx}{x^3(x^4 + a^4)} = -\frac{1}{2a^4 x^2} - \frac{1}{2a^6} \operatorname{tg}^{-1} \frac{x^2}{a^2}$$

$$1.15.8 \quad \int \frac{dx}{x^4 - a^4} = \frac{1}{4a^3} \ln \left(\frac{x - a}{x + a} \right) - \frac{1}{2a^3} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.15.9 \quad \int \frac{x dx}{x^4 - a^4} = \frac{1}{4a^2} \ln \left(\frac{x^2 - a^2}{x^2 + a^2} \right)$$

$$1.15.10 \quad \int \frac{x^2 dx}{x^4 - a^4} = \frac{1}{4a} \ln \left(\frac{x - a}{x + a} \right) + \frac{1}{2a} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.15.11 \quad \int \frac{x^3 dx}{x^4 - a^4} = \frac{1}{4} \ln(x^4 - a^4)$$

$$1.15.12 \quad \int \frac{dx}{x(x^4 - a^4)} = \frac{1}{4a^4} \ln \left(\frac{x^4 - a^4}{x^4} \right)$$

$$1.15.13 \quad \int \frac{dx}{x^3(x^4 - a^4)} = \frac{1}{a^4 x} + \frac{1}{4a^5} \ln \left(\frac{x - a}{x + a} \right) + \frac{1}{2a^5} \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.15.14 \quad \int \frac{dx}{x^3(x^4 - a^4)} = \frac{1}{2a^4 x^2} + \frac{1}{4a^6} \ln \left(\frac{x^2 - a^2}{x^2 + a^2} \right)$$

(16)

INTEGRALES QUE CONTIENEN $x^n \pm a^n$

$$1.16.1 \quad \int \frac{dx}{x(x^n + a^n)} = \frac{1}{na^n} \ln \frac{x^n}{x^n + a^n}$$

$$1.16.2 \quad \int \frac{x^{n-1} dx}{x^n + a^n} = \frac{1}{n} \ln(x^n + a^n)$$

$$1.16.3 \quad \int \frac{x^m dx}{(x^n + a^n)^r} = \int \frac{x^{m-n} dx}{(x^n + a^n)^{r-1}} - a^n \int \frac{x^{m-n} dx}{(x^n + a^n)^r}$$

$$1.16.4 \quad \int \frac{dx}{x^m(x^n + a^n)^r} = \frac{1}{a^n} \int \frac{dx}{x^m(x^n + a^n)^{r-1}} - \frac{1}{a^n} \int \frac{dx}{x^{m-n}(x^n + a^n)^r}$$

$$1.16.5 \quad \int \frac{dx}{x\sqrt{x^n + a^n}} = \frac{1}{n\sqrt{a^n}} \ln \left(\frac{\sqrt{x^n + a^n} - \sqrt{a^n}}{\sqrt{x^n + a^n} + \sqrt{a^n}} \right)$$

$$1.16.6 \quad \int \frac{dx}{x(x^n - a^n)} = \frac{1}{na^n} \ln \left(\frac{x^n - a^n}{x^n} \right)$$

$$1.16.7 \quad \int \frac{x^{n-1} dx}{x^n - a^n} = \frac{1}{n} \ln(x^n - a^n)$$

$$1.16.8 \quad \int \frac{x^m dx}{(x^n - a^n)^r} = a^n \int \frac{x^{m-n} dx}{(x^n - a^n)^r} + \int \frac{x^{m-n} dx}{(x^n - a^n)^{r-1}}$$

$$1.16.9 \quad \int \frac{dx}{x^m(x^n - a^n)^r} = \frac{1}{a^n} \int \frac{dx}{x^m(x^n - a^n)^{r-1}} - \frac{1}{a^n} \int \frac{dx}{x^{m-n}(x^n - a^n)^r}$$

$$1.16.10 \quad \int \frac{dx}{x\sqrt{x^n - a^n}} = \frac{2}{n\sqrt{a^n}} \cos^{-1} \sqrt{\frac{a^n}{x^n}}$$

$$1.16.11 \quad \int \frac{x^{p-1} dx}{x^{2m} + a^{2m}} = \frac{1}{ma^{2m-p}} \sum_{k=1}^m \operatorname{sen} \frac{(2k-1)p\pi}{2m} \operatorname{tg}^{-1} \left(\frac{x + a \cos[(2k-1)\pi/2m]}{a \operatorname{sen}[(2k-1)\pi/2m]} \right) \\ - \frac{1}{2ma^{2m-p}} \sum_{k=1}^m \cos \frac{(2k-1)p\pi}{2m} \ln \left(x^2 + 2ax \cos \frac{(2k-1)\pi}{2m} + a^2 \right)$$

donde $0 < p \leq 2m$.

$$1.16.12 \quad \int \frac{x^{p-1} dx}{x^{2m} - a^{2m}} = \frac{1}{2ma^{2m-p}} \sum_{k=1}^{m-1} \cos \frac{kp\pi}{m} \ln \left(x^2 - 2ax \cos \frac{k\pi}{m} + a^2 \right) \\ - \frac{1}{ma^{2m-p}} \sum_{k=1}^{m-1} \operatorname{sen} \frac{kp\pi}{m} \operatorname{tg}^{-1} \left(\frac{x - a \cos(k\pi/m)}{a \operatorname{sen}(k\pi/m)} \right) \\ + \frac{1}{2ma^{2m-p}} \{ \ln(x - a) + (-1)^p \ln(x + a) \}$$

donde $0 < p \leq 2m$.

$$\begin{aligned}
 1.16.13 \quad \int \frac{x^{p-1} dx}{x^{2m+1} + a^{2m+1}} &= \frac{2(-1)^{p-1}}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \operatorname{sen} \frac{2kp\pi}{2m+1} \operatorname{tg}^{-1} \left(\frac{x + a \cos [2k\pi/(2m+1)]}{a \operatorname{sen} [2k\pi/(2m+1)]} \right) \\
 &- \frac{(-1)^{p-1}}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \cos \frac{2kp\pi}{2m+1} \ln \left(x^2 + 2ax \cos \frac{2k\pi}{2m+1} + a^2 \right) \\
 &+ \frac{(-1)^{p-1} \ln(x+a)}{(2m+1)a^{2m-p+1}}
 \end{aligned}$$

donde $0 < p \leq 2m+1$.

$$\begin{aligned}
 1.16.14 \quad \int \frac{x^{p-1} dx}{x^{2m+1} - a^{2m+1}} &= \frac{-2}{(2m-1)a^{2m-p+1}} \sum_{k=1}^m \operatorname{sen} \frac{2kp\pi}{2m+1} \operatorname{tg}^{-1} \left(\frac{x - a \cos [2k\pi/(2m+1)]}{a \operatorname{sen} [2k\pi/(2m+1)]} \right) \\
 &+ \frac{1}{(2m+1)a^{2m-p+1}} \sum_{k=1}^m \cos \frac{2kp\pi}{2m+1} \ln \left(x^2 - 2ax \cos \frac{2k\pi}{2m+1} + a^2 \right) \\
 &+ \frac{\ln(x-a)}{(2m+1)a^{2m-p+1}}
 \end{aligned}$$

donde $0 < p \leq 2m+1$.

(17) INTEGRALES QUE CONTIENEN $\operatorname{sen} ax$

$$1.17.1 \quad \int \operatorname{sen} ax \, dx = -\frac{\cos ax}{a}$$

$$1.17.2 \quad \int x \operatorname{sen} ax \, dx = -\frac{\operatorname{sen} ax}{a^2} - \frac{x \cos ax}{a}$$

$$1.17.3 \quad \int x^2 \operatorname{sen} ax \, dx = \frac{2x}{a^2} \operatorname{sen} ax + \left(\frac{2}{a^3} - \frac{x^2}{a} \right) \cos ax$$

$$1.17.4 \quad \int x^3 \operatorname{sen} ax \, dx = \left(\frac{3x^2}{a^2} - \frac{6}{a^4} \right) \operatorname{sen} ax + \left(\frac{6x}{a^3} - \frac{x^3}{a} \right) \cos ax$$

$$1.17.5 \quad \int \frac{\operatorname{sen} ax}{x} dx = ax - \frac{(ax)^3}{3 \cdot 3!} + \frac{(ax)^5}{5 \cdot 5!} - \dots$$

$$1.17.6 \quad \int \frac{\operatorname{sen} ax}{x^2} dx = -\frac{\operatorname{sen} ax}{x} + a \int \frac{\cos ax}{x} dx \quad [\text{Véase 1.18.5}].$$

$$1.17.7 \quad \int \frac{dx}{\operatorname{sen} ax} = \frac{1}{a} \ln(\csc ax - \cot ax) = \frac{1}{a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.17.8 \quad \int \frac{x \, dx}{\operatorname{sen} ax} = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(2^{2n-1} - 1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.17.9 \quad \int \operatorname{sen}^2 ax \, dx = \frac{x}{2} - \frac{\operatorname{sen} 2ax}{4a}$$

$$1.17.10 \quad \int x \operatorname{sen}^2 ax \, dx = \frac{x^2}{4} - \frac{x \operatorname{sen} 2ax}{4a} - \frac{\cos 2ax}{8a^2}$$

$$1.17.11 \quad \int \operatorname{sen}^3 ax \, dx = -\frac{\cos ax}{a} + \frac{\cos^3 ax}{3a}$$

$$1.17.12 \quad \int \operatorname{sen}^4 ax \, dx = \frac{3x}{8} - \frac{\operatorname{sen} 2ax}{4a} + \frac{\operatorname{sen} 4ax}{32a}$$

$$1.17.13 \quad \int \frac{dx}{\operatorname{sen}^2 ax} = -\frac{1}{a} \cot ax$$

$$1.17.14 \quad \int \frac{dx}{\operatorname{sen}^3 ax} = -\frac{\cos ax}{2a \operatorname{sen}^2 ax} + \frac{1}{2a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.17.15 \quad \int \operatorname{sen} px \operatorname{sen} qx \, dx = \frac{\operatorname{sen}(p-q)x}{2(p-q)} - \frac{\operatorname{sen}(p+q)x}{2(p+q)} \quad [\text{Si } p = \pm q, \text{ véase 1.17.9}].$$

$$1.17.16 \quad \int \frac{dx}{1 - \operatorname{sen} ax} = \frac{1}{a} \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right)$$

$$1.17.17 \quad \int \frac{x \, dx}{1 - \operatorname{sen} ax} = \frac{x}{a} \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right) + \frac{2}{a^2} \ln \operatorname{sen} \left(\frac{\pi}{4} - \frac{ax}{2} \right)$$

$$1.17.18 \quad \int \frac{dx}{1 + \operatorname{sen} ax} = -\frac{1}{a} \operatorname{tg} \left(\frac{\pi}{4} - \frac{ax}{2} \right)$$

$$1.17.19 \quad \int \frac{x \, dx}{1 + \operatorname{sen} ax} = -\frac{x}{a} \operatorname{tg} \left(\frac{\pi}{4} - \frac{ax}{2} \right) + \frac{2}{a^2} \ln \operatorname{sen} \left(\frac{\pi}{4} + \frac{ax}{2} \right)$$

$$1.17.20 \quad \int \frac{dx}{(1 - \operatorname{sen} ax)^2} = \frac{1}{2a} \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right) + \frac{1}{6a} \operatorname{tg}^3 \left(\frac{\pi}{4} + \frac{ax}{2} \right)$$

$$1.17.21 \quad \int \frac{dx}{(1 + \operatorname{sen} ax)^2} = -\frac{1}{2a} \operatorname{tg} \left(\frac{\pi}{4} - \frac{ax}{2} \right) - \frac{1}{6a} \operatorname{tg}^3 \left(\frac{\pi}{4} - \frac{ax}{2} \right)$$

$$1.17.22 \quad \int \frac{dx}{p + q \operatorname{sen} ax} = \begin{cases} \frac{2}{a \sqrt{p^2 - q^2}} \operatorname{tg}^{-1} \frac{p \operatorname{tg} \frac{1}{2} ax + q}{\sqrt{p^2 - q^2}} \\ \frac{1}{a \sqrt{q^2 - p^2}} \ln \left(\frac{p \operatorname{tg} \frac{1}{2} ax + q - \sqrt{q^2 - p^2}}{p \operatorname{tg} \frac{1}{2} ax + q + \sqrt{q^2 - p^2}} \right) \end{cases}$$

Si $p = \pm q$, véase 1.17.16 y 1.17.18.

$$1.17.23 \quad \int \frac{dx}{(p + q \operatorname{sen} ax)^2} = \frac{q \cos ax}{a(p^2 - q^2)(p + q \operatorname{sen} ax)} + \frac{p}{p^2 - q^2} \int \frac{dx}{p + q \operatorname{sen} ax}$$

Si $p = \pm q$, véase 1.17.20 y 1.17.21.

$$1.17.24 \quad \int \frac{dx}{p^2 + q^2 \operatorname{sen}^2 ax} = \frac{1}{ap\sqrt{p^2 + q^2}} \operatorname{tg}^{-1} \frac{\sqrt{p^2 + q^2} \operatorname{tg} ax}{p}$$

$$1.17.25 \quad \int \frac{dx}{p^2 - q^2 \operatorname{sen}^2 ax} = \begin{cases} \frac{1}{ap\sqrt{p^2 - q^2}} \operatorname{tg}^{-1} \frac{\sqrt{p^2 - q^2} \operatorname{tg} ax}{p} \\ \frac{1}{2ap\sqrt{q^2 - p^2}} \ln \left(\frac{\sqrt{q^2 - p^2} \operatorname{tg} ax + p}{\sqrt{q^2 - p^2} \operatorname{tg} ax - p} \right) \end{cases}$$

$$1.17.26 \quad \int x^m \operatorname{sen} ax \, dx = -\frac{x^m \cos ax}{a} + \frac{mx^{m-1} \operatorname{sen} ax}{a^2} - \frac{m(m-1)}{a^2} \int x^{m-2} \operatorname{sen} ax \, dx$$

$$1.17.27 \quad \int \frac{\operatorname{sen} ax}{x^n} dx = -\frac{\operatorname{sen} ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\cos ax}{x^{n-1}} dx \quad [\text{Véase 1.18.30}].$$

$$1.17.28 \quad \int \operatorname{sen}^n ax \, dx = -\frac{\operatorname{sen}^{n-1} ax \cos ax}{an} + \frac{n-1}{n} \int \operatorname{sen}^{n-2} ax \, dx$$

$$1.17.29 \quad \int \frac{dx}{\operatorname{sen}^n ax} = \frac{-\cos ax}{a(n-1)\operatorname{sen}^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\operatorname{sen}^{n-2} ax}$$

$$1.17.30 \quad \int \frac{x \, dx}{\operatorname{sen}^n ax} = \frac{-x \cos ax}{a(n-1)\operatorname{sen}^{n-1} ax} - \frac{1}{a^2(n-1)(n-2)\operatorname{sen}^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x \, dx}{\operatorname{sen}^{n-2} ax}$$

(18) INTEGRALES QUE CONTIENEN $\cos ax$

$$1.18.1 \quad \int \cos ax \, dx = \frac{\operatorname{sen} ax}{a}$$

$$1.18.2 \quad \int x \cos ax \, dx = \frac{\cos ax}{a^2} + \frac{x \operatorname{sen} ax}{a}$$

$$1.18.3 \quad \int x^2 \cos ax \, dx = \frac{2x}{a^2} \cos ax + \left(\frac{x^2}{a} - \frac{2}{a^3} \right) \operatorname{sen} ax$$

$$1.18.4 \quad \int x^3 \cos ax \, dx = \left(\frac{3x^2}{a^2} - \frac{6}{a^4} \right) \cos ax + \left(\frac{x^3}{a} - \frac{6x}{a^3} \right) \operatorname{sen} ax$$

$$1.18.5 \quad \int \frac{\cos ax}{x} dx = \ln x - \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^4}{4 \cdot 4!} - \frac{(ax)^6}{6 \cdot 6!} + \dots$$

$$1.18.6 \quad \int \frac{\cos ax}{x^2} dx = -\frac{\cos ax}{x} - a \int \frac{\operatorname{sen} ax}{x} dx \quad [\text{Véase 1.17.5}].$$

$$1.18.7 \quad \int \frac{dx}{\cos ax} = \frac{1}{a} \ln(\sec ax + \operatorname{tg} ax) = \frac{1}{a} \ln \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right)$$

$$1.18.8 \quad \int \frac{x dx}{\cos ax} = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} + \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{E_n(ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$1.18.9 \quad \int \cos^2 ax dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$$

$$1.18.10 \quad \int x \cos^2 ax dx = \frac{x^2}{4} + \frac{x \sin 2ax}{4a} + \frac{\cos 2ax}{8a^2}$$

$$1.18.11 \quad \int \cos^3 ax dx = \frac{\sin ax}{a} - \frac{\sin^3 ax}{3a}$$

$$1.18.12 \quad \int \cos^4 ax dx = \frac{3x}{8} + \frac{\sin 2ax}{4a} + \frac{\sin 4ax}{32a}$$

$$1.18.13 \quad \int \frac{dx}{\cos^2 ax} = \frac{\operatorname{tg} ax}{a}$$

$$1.18.14 \quad \int \frac{dx}{\cos^3 ax} = \frac{\sin ax}{2a \cos^2 ax} + \frac{1}{2a} \ln \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right)$$

$$1.18.15 \quad \int \cos ax \cos px dx = \frac{\sin(a-p)x}{2(a-p)} + \frac{\sin(a+p)x}{2(a+p)} \quad [\text{Si } a = \pm p, \text{ véase 1.18.9}].$$

$$1.18.16 \quad \int \frac{dx}{1 - \cos ax} = -\frac{1}{a} \cot \frac{ax}{2}$$

$$1.18.17 \quad \int \frac{x dx}{1 - \cos ax} = -\frac{x}{a} \cot \frac{ax}{2} + \frac{2}{a^2} \ln \sin \frac{ax}{2}$$

$$1.18.18 \quad \int \frac{dx}{1 + \cos ax} = \frac{1}{a} \operatorname{tg} \frac{ax}{2}$$

$$1.18.19 \quad \int \frac{x dx}{1 + \cos ax} = \frac{x}{a} \operatorname{tg} \frac{ax}{2} + \frac{2}{a^2} \ln \cos \frac{ax}{2}$$

$$1.18.20 \quad \int \frac{dx}{(1 - \cos ax)^2} = -\frac{1}{2a} \cot \frac{ax}{2} - \frac{1}{6a} \cot^3 \frac{ax}{2}$$

$$1.18.21 \quad \int \frac{dx}{(1 + \cos ax)^2} = \frac{1}{2a} \operatorname{tg} \frac{ax}{2} + \frac{1}{6a} \operatorname{tg}^3 \frac{ax}{2}$$

$$1.18.22 \quad \int \frac{dx}{p + q \cos ax} = \begin{cases} \frac{2}{a\sqrt{p^2 - q^2}} \operatorname{tg}^{-1} \sqrt{(p-q)/(p+q)} \operatorname{tg} \frac{1}{2} ax \\ \frac{1}{a\sqrt{q^2 - p^2}} \ln \left(\frac{\operatorname{tg} \frac{1}{2} ax + \sqrt{(q+p)/(q-p)}}{\operatorname{tg} \frac{1}{2} ax - \sqrt{(q+p)/(q-p)}} \right) \end{cases} \quad [\text{Si } p = \pm q, \text{ véanse 1.18.16 y 1.18.18}].$$

$$1.18.23 \quad \int \frac{dx}{(p + q \cos ax)^2} = \frac{q \sin ax}{a(q^2 - p^2)(p + q \cos ax)} - \frac{p}{q^2 - p^2} \int \frac{dx}{p + q \cos ax} \quad [\text{Si } p = \pm q, \text{ véanse 1.18.19 y 1.18.20}].$$

$$1.18.24 \quad \int \frac{dx}{p^2 + q^2 \cos^2 ax} = \frac{1}{ap \sqrt{p^2 + q^2}} \operatorname{tg}^{-1} \frac{p \operatorname{tg} ax}{\sqrt{p^2 + q^2}}$$

$$1.18.25 \quad \int \frac{dx}{p^2 - q^2 \cos^2 ax} = \begin{cases} \frac{1}{ap \sqrt{p^2 - q^2}} \operatorname{tg}^{-1} \frac{p \operatorname{tg} ax}{\sqrt{p^2 - q^2}} \\ \frac{1}{2ap \sqrt{q^2 - p^2}} \ln \left(\frac{p \operatorname{tg} ax - \sqrt{q^2 - p^2}}{p \operatorname{tg} ax + \sqrt{q^2 - p^2}} \right) \end{cases}$$

$$1.18.26 \quad \int x^m \cos ax \, dx = \frac{x^m \operatorname{sen} ax}{a} + \frac{mx^{m-1}}{a^2} \cos ax - \frac{m(m-1)}{a^2} \int x^{m-2} \cos ax \, dx$$

$$1.18.27 \quad \int \frac{\cos ax}{x^n} dx = -\frac{\cos ax}{(n-1)x^{n-1}} - \frac{a}{n-1} \int \frac{\operatorname{sen} ax}{x^{n-1}} dx \quad [\text{Véase 1.17.27}].$$

$$1.18.28 \quad \int \cos^n ax \, dx = \frac{\operatorname{sen} ax \cos^{n-1} ax}{an} + \frac{n-1}{n} \int \cos^{n-2} ax \, dx$$

$$1.18.29 \quad \int \frac{dx}{\cos^n ax} = \frac{\operatorname{sen} ax}{a(n-1) \cos^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\cos^{n-2} ax}$$

$$1.18.30 \quad \int \frac{x \, dx}{\cos^n ax} = \frac{x \operatorname{sen} ax}{a(n-1) \cos^{n-1} ax} - \frac{1}{a^2(n-1)(n-2) \cos^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x \, dx}{\cos^{n-2} ax}$$

(19) INTEGRALES QUE CONTIENEN $\operatorname{sen} ax$ Y $\cos ax$

$$1.19.1 \quad \int \operatorname{sen} ax \cos ax \, dx = \frac{\operatorname{sen}^2 ax}{2a}$$

$$1.19.2 \quad \int \operatorname{sen} px \cos qx \, dx = -\frac{\cos(p-q)x}{2(p-q)} - \frac{\cos(p+q)x}{2(p+q)}$$

$$1.19.3 \quad \int \operatorname{sen}^n ax \cos ax \, dx = \frac{\operatorname{sen}^{n+1} ax}{(n+1)a} \quad [\text{Si } n = -1, \text{ véase 1.21.1}].$$

$$1.19.4 \quad \int \cos^n ax \operatorname{sen} ax \, dx = -\frac{\cos^{n+1} ax}{(n+1)a} \quad [\text{Si } n = -1, \text{ véase 1.20.1}].$$

$$1.19.5 \quad \int \operatorname{sen}^2 ax \cos^2 ax \, dx = \frac{x}{8} - \frac{\operatorname{sen} 4ax}{32a}$$

$$1.19.6 \quad \int \frac{dx}{\operatorname{sen} ax \cos ax} = \frac{1}{a} \ln \operatorname{tg} ax$$

$$1.19.7 \quad \int \frac{dx}{\operatorname{sen}^2 ax \cos ax} = \frac{1}{a} \ln \operatorname{tg} \left(\frac{\pi}{4} + \frac{ax}{2} \right) - \frac{1}{a \operatorname{sen} ax}$$

$$1.19.8 \quad \int \frac{dx}{\operatorname{sen} ax \cos^2 ax} = \frac{1}{a} \ln \operatorname{tg} \frac{ax}{2} + \frac{1}{a \cos ax}$$

$$1.19.9 \quad \int \frac{dx}{\sin^2 ax \cos^2 ax} = -\frac{2 \cot 2ax}{a}$$

$$1.19.10 \quad \int \frac{\sin^2 ax}{\cos ax} dx = -\frac{\sin ax}{a} + \frac{1}{a} \ln \operatorname{tg} \left(\frac{ax}{2} + \frac{\pi}{4} \right)$$

$$1.19.11 \quad \int \frac{\cos^2 ax}{\sin ax} dx = \frac{\cos ax}{a} + \frac{1}{a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.19.12 \quad \int \frac{dx}{\cos ax(1 \pm \sin ax)} = \mp \frac{1}{2a(1 \pm \sin ax)} + \frac{1}{2a} \ln \operatorname{tg} \left(\frac{ax}{2} + \frac{\pi}{4} \right)$$

$$1.19.13 \quad \int \frac{dx}{\sin ax(1 \pm \cos ax)} = \pm \frac{1}{2a(1 \pm \cos ax)} + \frac{1}{2a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.19.14 \quad \int \frac{dx}{\sin ax \pm \cos ax} = \frac{1}{a\sqrt{2}} \ln \operatorname{tg} \left(\frac{ax}{2} \pm \frac{\pi}{8} \right)$$

$$1.19.15 \quad \int \frac{\sin ax \, dx}{\sin ax \pm \cos ax} = \frac{x}{2} \mp \frac{1}{2a} \ln(\sin ax \pm \cos ax)$$

$$1.19.16 \quad \int \frac{\cos ax \, dx}{\sin ax \pm \cos ax} = \pm \frac{x}{2} + \frac{1}{2a} \ln(\sin ax \pm \cos ax)$$

$$1.19.17 \quad \int \frac{\sin ax \, dx}{p + q \cos ax} = -\frac{1}{aq} \ln(p + q \cos ax)$$

$$1.19.18 \quad \int \frac{\cos ax \, dx}{p + q \sin ax} = \frac{1}{aq} \ln(p + q \sin ax)$$

$$1.19.19 \quad \int \frac{\sin ax \, dx}{(p + q \cos ax)^n} = \frac{1}{aq(n-1)(p + q \cos ax)^{n-1}}$$

$$1.19.20 \quad \int \frac{\cos ax \, dx}{(p + q \sin ax)^n} = \frac{-1}{aq(n-1)(p + q \sin ax)^{n-1}}$$

$$1.19.21 \quad \int \frac{dx}{p \sin ax + q \cos ax} = \frac{1}{a\sqrt{p^2 + q^2}} \ln \operatorname{tg} \left(\frac{ax + \operatorname{tg}^{-1}(q/p)}{2} \right)$$

$$1.19.22 \quad \int \frac{dx}{p \sin ax + q \cos ax + r} = \begin{cases} \frac{2}{a\sqrt{r^2 - p^2 - q^2}} \operatorname{tg}^{-1} \left(\frac{p + (r-q) \operatorname{tg}(ax/2)}{\sqrt{r^2 - p^2 - q^2}} \right) \\ \frac{1}{a\sqrt{p^2 + q^2 - r^2}} \ln \left(\frac{p - \sqrt{p^2 + q^2 - r^2} + (r-q) \operatorname{tg}(ax/2)}{p + \sqrt{p^2 + q^2 - r^2} + (r-q) \operatorname{tg}(ax/2)} \right) \end{cases}$$

Si $r = q$, véase 1.19.23. Si $r^2 = p^2 + q^2$, véase 1.19.24.

$$1.19.23 \quad \int \frac{dx}{p \sin ax + q(1 + \cos ax)} = \frac{1}{ap} \ln \left(q + p \operatorname{tg} \frac{ax}{2} \right)$$

$$1.19.24 \quad \int \frac{dx}{p \sin ax + q \cos ax \pm \sqrt{p^2 + q^2}} = \frac{-1}{a\sqrt{p^2 + q^2}} \operatorname{tg} \left(\frac{\pi}{4} \mp \frac{ax + \operatorname{tg}^{-1}(q/p)}{2} \right)$$

$$1.19.25 \quad \int \frac{dx}{p^2 \sin^2 ax + q^2 \cos^2 ax} = \frac{1}{apq} \operatorname{tg}^{-1} \left(\frac{p \operatorname{tg} ax}{q} \right)$$

$$1.19.26 \quad \int \frac{dx}{p^2 \sin^2 ax - q^2 \cos^2 ax} = \frac{1}{2apq} \ln \left(\frac{p \operatorname{tg} ax - q}{p \operatorname{tg} ax + q} \right)$$

$$1.19.27 \quad \int \sin^m ax \cos^n ax \, dx = \begin{cases} -\frac{\sin^{m-1} ax \cos^{n+1} ax}{a(m+n)} + \frac{m-1}{m+n} \int \sin^{m-2} ax \cos^n ax \, dx \\ \frac{\sin^{m+1} ax \cos^{n-1} ax}{a(m+n)} + \frac{n-1}{m+n} \int \sin^m ax \cos^{n-2} ax \, dx \end{cases}$$

$$1.19.28 \quad \int \frac{\sin^m ax}{\cos^n ax} \, dx = \begin{cases} \frac{\sin^{m-1} ax}{a(n-1) \cos^{n-1} ax} - \frac{m-1}{n-1} \int \frac{\sin^{m-2} ax}{\cos^{n-2} ax} \, dx \\ \frac{\sin^{m+1} ax}{a(n-1) \cos^{n-1} ax} - \frac{m-n+2}{n-1} \int \frac{\sin^m ax}{\cos^{n-2} ax} \, dx \\ \frac{-\sin^{m-1} ax}{a(m-n) \cos^{n-1} ax} + \frac{m-1}{m-n} \int \frac{\sin^{m-2} ax}{\cos^n ax} \, dx \end{cases}$$

$$1.19.29 \quad \int \frac{\cos^m ax}{\sin^n ax} \, dx = \begin{cases} \frac{-\cos^{m-1} ax}{a(n-1) \sin^{n-1} ax} - \frac{m-1}{n-1} \int \frac{\cos^{m-2} ax}{\sin^{n-2} ax} \, dx \\ \frac{-\cos^{m+1} ax}{a(n-1) \sin^{n-1} ax} - \frac{m-n+2}{n-1} \int \frac{\cos^m ax}{\sin^{n-2} ax} \, dx \\ \frac{\cos^{m-1} ax}{a(m-n) \sin^{n-1} ax} + \frac{m-1}{m-n} \int \frac{\cos^{m-2} ax}{\sin^n ax} \, dx \end{cases}$$

$$1.19.30 \quad \int \frac{dx}{\sin^m ax \cos^n ax} = \begin{cases} \frac{1}{a(n-1) \sin^{m-1} ax \cos^{n-1} ax} + \frac{m+n-2}{n-1} \int \frac{dx}{\sin^m ax \cos^{n-2} ax} \\ \frac{-1}{a(m-1) \sin^{m-1} ax \cos^{n-1} ax} + \frac{m+n-2}{m-1} \int \frac{dx}{\sin^{m-2} ax \cos^n ax} \end{cases}$$

(20) INTEGRALES QUE CONTIENEN $\operatorname{tg} ax$

$$1.20.1 \quad \int \operatorname{tg} ax \, dx = -\frac{1}{a} \ln \cos ax = \frac{1}{a} \ln \sec ax$$

$$1.20.2 \quad \int \operatorname{tg}^2 ax \, dx = \frac{\operatorname{tg} ax}{a} - x$$

$$1.20.3 \quad \int \operatorname{tg}^3 ax \, dx = \frac{\operatorname{tg}^2 ax}{2a} + \frac{1}{a} \ln \cos ax$$

$$1.20.4 \quad \int \operatorname{tg}^n ax \sec^2 ax \, dx = \frac{\operatorname{tg}^{n+1} ax}{(n+1)a}$$

$$1.20.5 \quad \int \frac{\sec^2 ax}{\operatorname{tg} ax} dx = \frac{1}{a} \ln \operatorname{tg} ax$$

$$1.20.6 \quad \int \frac{dx}{\operatorname{tg} ax} = \frac{1}{a} \ln \operatorname{sen} ax$$

$$1.20.7 \quad \int x \operatorname{tg} ax dx = \frac{1}{a^2} \left\{ \frac{(ax)^3}{3} + \frac{(ax)^5}{15} + \frac{2(ax)^7}{105} + \dots + \frac{2^{2n}(2^{2n}-1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.20.8 \quad \int \frac{\operatorname{tg} ax}{x} dx = ax + \frac{(ax)^3}{9} + \frac{2(ax)^5}{75} + \dots + \frac{2^{2n}(2^{2n}-1)B_n(ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$1.20.9 \quad \int x \operatorname{tg}^2 ax dx = \frac{x \operatorname{tg} ax}{a} + \frac{1}{a^2} \ln \cos ax - \frac{x^2}{2}$$

$$1.20.10 \quad \int \frac{dx}{p + q \operatorname{tg} ax} = \frac{px}{p^2 + q^2} + \frac{q}{a(p^2 + q^2)} \ln(q \operatorname{sen} ax + p \cos ax)$$

$$1.20.11 \quad \int \operatorname{tg}^n ax dx = \frac{\operatorname{tg}^{n-1} ax}{(n-1)a} - \int \operatorname{tg}^{n-2} ax dx$$

(21)

INTEGRALES QUE CONTIENEN $\cot ax$

$$1.21.1 \quad \int \cot ax dx = \frac{1}{a} \ln \operatorname{sen} ax$$

$$1.21.2 \quad \int \cot^2 ax dx = -\frac{\cot ax}{a} - x$$

$$1.21.3 \quad \int \cot^3 ax dx = -\frac{\cot^2 ax}{2a} - \frac{1}{a} \ln \operatorname{sen} ax$$

$$1.21.4 \quad \int \cot^n ax \csc^2 ax dx = -\frac{\cot^{n+1} ax}{(n+1)a}$$

$$1.21.5 \quad \int \frac{\csc^2 ax}{\cot ax} dx = -\frac{1}{a} \ln \cot ax$$

$$1.21.6 \quad \int \frac{dx}{\cot ax} = -\frac{1}{a} \ln \cos ax$$

$$1.21.7 \quad \int x \cot ax dx = \frac{1}{a^2} \left\{ ax - \frac{(ax)^3}{9} - \frac{(ax)^5}{225} - \dots - \frac{2^{2n}B_n(ax)^{2n+1}}{(2n+1)!} - \dots \right\}$$

$$1.21.8 \quad \int \frac{\cot ax}{x} dx = -\frac{1}{ax} - \frac{ax}{3} - \frac{(ax)^3}{135} - \dots - \frac{2^{2n}B_n(ax)^{2n-1}}{(2n-1)(2n)!} - \dots$$

$$1.21.9 \quad \int x \cot^2 ax dx = -\frac{x \cot ax}{a} + \frac{1}{a^2} \ln \operatorname{sen} ax - \frac{x^2}{2}$$

$$1.21.10 \quad \int \frac{dx}{p + q \cot ax} = \frac{px}{p^2 + q^2} - \frac{q}{a(p^2 + q^2)} \ln(q \sin ax + q \cos ax)$$

$$1.21.11 \quad \int \cot^n ax \, dx = -\frac{\cot^{n-1} ax}{(n-1)a} - \int \cot^{n-2} ax \, dx$$

(22) INTEGRALES QUE CONTIENEN $\sec ax$

$$1.22.1 \quad \int \sec ax \, dx = \frac{1}{a} \ln(\sec ax + \operatorname{tg} ax) = \frac{1}{a} \ln \operatorname{tg} \left(\frac{ax}{2} + \frac{\pi}{4} \right)$$

$$1.22.2 \quad \int \sec^2 ax \, dx = \frac{\operatorname{tg} ax}{a}$$

$$1.22.3 \quad \int \sec^3 ax \, dx = \frac{\sec ax \operatorname{tg} ax}{2a} + \frac{1}{2a} \ln(\sec ax + \operatorname{tg} ax)$$

$$1.22.4 \quad \int \sec^n ax \operatorname{tg} ax \, dx = \frac{\sec^n ax}{na}$$

$$1.22.5 \quad \int \frac{dx}{\sec ax} = \frac{\sin ax}{a}$$

$$1.22.6 \quad \int x \sec ax \, dx = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} + \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{E_n(ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$1.22.7 \quad \int \frac{\sec ax}{x} \, dx = \ln x + \frac{(ax)^2}{4} + \frac{5(ax)^4}{96} + \frac{61(ax)^6}{4320} + \dots + \frac{E_n(ax)^{2n}}{2n(2n)!} + \dots$$

$$1.22.8 \quad \int x \sec^2 ax \, dx = \frac{x}{a} \operatorname{tg} ax + \frac{1}{a^2} \ln \cos ax$$

$$1.22.9 \quad \int \frac{dx}{q + p \sec ax} = \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \cos ax}$$

$$1.22.10 \quad \int \sec^n ax \, dx = \frac{\sec^{n-2} ax \operatorname{tg} ax}{a(n-1)} + \frac{n-2}{n-1} \int \sec^{n-2} ax \, dx$$

(23) INTEGRALES QUE CONTIENEN $\csc ax$

$$1.23.1 \quad \int \csc ax \, dx = \frac{1}{a} \ln(\csc ax - \cot ax) = \frac{1}{a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.23.2 \quad \int \csc^2 ax \, dx = -\frac{\cot ax}{a}$$

$$1.23.3 \quad \int \csc^3 ax \, dx = -\frac{\csc ax \cot ax}{2a} + \frac{1}{2a} \ln \operatorname{tg} \frac{ax}{2}$$

$$1.23.4 \quad \int \csc^n ax \cot ax \, dx = -\frac{\csc^n ax}{na}$$

$$1.23.5 \quad \int \frac{dx}{\csc ax} = -\frac{\cos ax}{a}$$

$$1.23.6 \quad \int x \csc ax \, dx = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(2^{2n-1} - 1)B_n(ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.23.7 \quad \int \frac{\csc ax}{x} \, dx = -\frac{1}{ax} + \frac{ax}{6} + \frac{7(ax)^3}{1080} + \dots + \frac{2(2^{2n-1} - 1)B_n(ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$1.23.8 \quad \int x \csc^2 ax \, dx = -\frac{x \cot ax}{a} + \frac{1}{a^2} \ln \operatorname{sen} ax$$

$$1.23.9 \quad \int \frac{dx}{q + p \csc ax} = \frac{x}{q} - \frac{p}{q} \int \frac{dx}{p + q \operatorname{sen} ax} \quad [\text{Véase 1.17.22}].$$

$$1.23.10 \quad \int \csc^n ax \, dx = -\frac{\csc^{n-2} ax \cot ax}{a(n-1)} + \frac{n-2}{n-1} \int \csc^{n-2} ax \, dx$$

(24) INTEGRALES QUE CONTIENEN FUNCIONES TRIGONOMÉTRICAS INVERSAS

$$1.24.1 \quad \int \operatorname{sen}^{-1} \frac{x}{a} \, dx = x \operatorname{sen}^{-1} \frac{x}{a} + \sqrt{a^2 - x^2}$$

$$1.24.2 \quad \int x \operatorname{sen}^{-1} \frac{x}{a} \, dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \operatorname{sen}^{-1} \frac{x}{a} + \frac{x \sqrt{a^2 - x^2}}{4}$$

$$1.24.3 \quad \int x^2 \operatorname{sen}^{-1} \frac{x}{a} \, dx = \frac{x^3}{3} \operatorname{sen}^{-1} \frac{x}{a} + \frac{(x^2 + 2a^2) \sqrt{a^2 - x^2}}{9}$$

$$1.24.4 \quad \int \frac{\operatorname{sen}^{-1}(x/a)}{x} \, dx = \frac{x}{a} + \frac{(x/a)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3(x/a)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5(x/a)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots$$

$$1.24.5 \quad \int \frac{\operatorname{sen}^{-1}(x/a)}{x^2} \, dx = -\frac{\operatorname{sen}^{-1}(x/a)}{x} - \frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.24.6 \quad \int \left(\operatorname{sen}^{-1} \frac{x}{a} \right)^2 \, dx = x \left(\operatorname{sen}^{-1} \frac{x}{a} \right)^2 - 2x + 2 \sqrt{a^2 - x^2} \operatorname{sen}^{-1} \frac{x}{a}$$

$$1.24.7 \quad \int \cos^{-1} \frac{x}{a} \, dx = x \cos^{-1} \frac{x}{a} - \sqrt{a^2 - x^2}$$

$$1.24.8 \quad \int x \cos^{-1} \frac{x}{a} \, dx = \left(\frac{x^2}{2} - \frac{a^2}{4} \right) \cos^{-1} \frac{x}{a} - \frac{x \sqrt{a^2 - x^2}}{4}$$

$$1.24.9 \quad \int x^2 \cos^{-1} \frac{x}{a} dx = \frac{x^3}{3} \cos^{-1} \frac{x}{a} - \frac{(x^2 + 2a^2) \sqrt{a^2 - x^2}}{9}$$

$$1.24.10 \quad \int \frac{\cos^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x - \int \frac{\operatorname{sen}^{-1}(x/a)}{x} dx \quad [\text{Véase 1.24.4}].$$

$$1.24.11 \quad \int \frac{\cos^{-1}(x/a)}{x^2} dx = -\frac{\cos^{-1}(x/a)}{x} + \frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right)$$

$$1.24.12 \quad \int \left(\cos^{-1} \frac{x}{a} \right)^2 dx = x \left(\cos^{-1} \frac{x}{a} \right)^2 - 2x - 2 \sqrt{a^2 - x^2} \cos^{-1} \frac{x}{a}$$

$$1.24.13 \quad \int \operatorname{tg}^{-1} \frac{x}{a} dx = x \operatorname{tg}^{-1} \frac{x}{a} - \frac{a}{2} \ln(x^2 + a^2)$$

$$1.24.14 \quad \int x \operatorname{tg}^{-1} \frac{x}{a} dx = \frac{1}{2} (x^2 + a^2) \operatorname{tg}^{-1} \frac{x}{a} - \frac{ax}{2}$$

$$1.24.15 \quad \int x^2 \operatorname{tg}^{-1} \frac{x}{a} dx = \frac{x^3}{3} \operatorname{tg}^{-1} \frac{x}{a} - \frac{ax^2}{6} + \frac{a^3}{6} \ln(x^2 + a^2)$$

$$1.24.16 \quad \int \frac{\operatorname{tg}^{-1}(x/a)}{x} dx = \frac{x}{a} - \frac{(x/a)^3}{3^2} + \frac{(x/a)^5}{5^2} - \frac{(x/a)^7}{7^2} + \dots$$

$$1.24.17 \quad \int \frac{\operatorname{tg}^{-1}(x/a)}{x^2} dx = -\frac{1}{x} \operatorname{tg}^{-1} \frac{x}{a} - \frac{1}{2a} \ln \left(\frac{x^2 + a^2}{x^2} \right)$$

$$1.24.18 \quad \int \cot^{-1} \frac{x}{a} dx = x \cot^{-1} \frac{x}{a} + \frac{a}{2} \ln(x^2 + a^2)$$

$$1.24.19 \quad \int x \cot^{-1} \frac{x}{a} dx = \frac{1}{2} (x^2 + a^2) \cot^{-1} \frac{x}{a} + \frac{ax}{2}$$

$$1.24.20 \quad \int x^2 \cot^{-1} \frac{x}{a} dx = \frac{x^3}{3} \cot^{-1} \frac{x}{a} + \frac{ax^2}{6} - \frac{a^3}{6} \ln(x^2 + a^2)$$

$$1.24.21 \quad \int \frac{\cot^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x - \int \frac{\operatorname{tg}^{-1}(x/a)}{x} dx \quad [\text{Véase 1.24.16}].$$

$$1.24.22 \quad \int \frac{\cot^{-1}(x/a)}{x^2} dx = -\frac{\cot^{-1}(x/a)}{x} + \frac{1}{2a} \ln \left(\frac{x^2 + a^2}{x^2} \right)$$

$$1.24.23 \quad \int \sec^{-1} \frac{x}{a} dx = \begin{cases} x \sec^{-1} \frac{x}{a} - a \ln(x + \sqrt{x^2 - a^2}) & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ x \sec^{-1} \frac{x}{a} + a \ln(x + \sqrt{x^2 - a^2}) & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$1.24.24 \quad \int x \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^2}{2} \sec^{-1} \frac{x}{a} - \frac{a\sqrt{x^2 - a^2}}{2} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^2}{2} \sec^{-1} \frac{x}{a} + \frac{a\sqrt{x^2 - a^2}}{2} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$1.24.25 \quad \int x^2 \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^3}{3} \sec^{-1} \frac{x}{a} - \frac{ax\sqrt{x^2 - a^2}}{6} - \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^3}{3} \sec^{-1} \frac{x}{a} + \frac{ax\sqrt{x^2 - a^2}}{6} + \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$1.24.26 \quad \int \frac{\sec^{-1}(x/a)}{x} dx = \frac{\pi}{2} \ln x + \frac{a}{x} + \frac{(a/x)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3(a/x)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5(a/x)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots$$

$$1.24.27 \quad \int \frac{\sec^{-1}(x/a)}{x^2} dx = \begin{cases} -\frac{\sec^{-1}(x/a)}{x} + \frac{\sqrt{x^2 - a^2}}{ax} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ -\frac{\sec^{-1}(x/a)}{x} - \frac{\sqrt{x^2 - a^2}}{ax} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$1.24.28 \quad \int \csc^{-1} \frac{x}{a} dx = \begin{cases} x \csc^{-1} \frac{x}{a} + a \ln(x + \sqrt{x^2 - a^2}) & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ x \csc^{-1} \frac{x}{a} - a \ln(x + \sqrt{x^2 - a^2}) & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$1.24.29 \quad \int x \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^2}{2} \csc^{-1} \frac{x}{a} + \frac{a\sqrt{x^2 - a^2}}{2} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^2}{2} \csc^{-1} \frac{x}{a} - \frac{a\sqrt{x^2 - a^2}}{2} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$1.24.30 \quad \int x^2 \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^3}{3} \csc^{-1} \frac{x}{a} + \frac{ax\sqrt{x^2 - a^2}}{6} + \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^3}{3} \csc^{-1} \frac{x}{a} - \frac{ax\sqrt{x^2 - a^2}}{6} - \frac{a^3}{6} \ln(x + \sqrt{x^2 - a^2}) & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$1.24.31 \quad \int \frac{\csc^{-1}(x/a)}{x} dx = -\left(\frac{a}{x} + \frac{(a/x)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3(a/x)^5}{2 \cdot 4 \cdot 5 \cdot 5} + \frac{1 \cdot 3 \cdot 5(a/x)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots\right)$$

$$1.24.32 \quad \int \frac{\csc^{-1}(x/a)}{x^2} dx = \begin{cases} -\frac{\csc^{-1}(x/a)}{x} - \frac{\sqrt{x^2 - a^2}}{ax} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ -\frac{\csc^{-1}(x/a)}{x} + \frac{\sqrt{x^2 - a^2}}{ax} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

$$1.24.33 \quad \int x^m \sec^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \sec^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{a^2 - x^2}} dx$$

$$1.24.34 \quad \int x^m \cos^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \cos^{-1} \frac{x}{a} + \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{a^2 - x^2}} dx$$

$$1.24.35 \quad \int x^m \operatorname{tg}^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \operatorname{tg}^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^{m+1}}{x^2 + a^2} dx$$

$$1.24.36 \quad \int x^m \cot^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \cot^{-1} \frac{x}{a} + \frac{a}{m+1} \int \frac{x^{m+1}}{x^2 + a^2} dx$$

$$1.24.37 \quad \int x^m \sec^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1} \sec^{-1}(x/a)}{m+1} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & 0 < \sec^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^{m+1} \sec^{-1}(x/a)}{m+1} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & \frac{\pi}{2} < \sec^{-1} \frac{x}{a} < \pi \end{cases}$$

$$1.24.38 \quad \int x^m \csc^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1} \csc^{-1}(x/a)}{m+1} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & 0 < \csc^{-1} \frac{x}{a} < \frac{\pi}{2} \\ \frac{x^{m+1} \csc^{-1}(x/a)}{m+1} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 - a^2}} & -\frac{\pi}{2} < \csc^{-1} \frac{x}{a} < 0 \end{cases}$$

(25) INTEGRALES QUE CONTIENEN e^{ax}

$$1.25.1 \quad \int e^{ax} dx = \frac{e^{ax}}{a}$$

$$1.25.2 \quad \int x e^{ax} dx = \frac{e^{ax}}{a} \left(x - \frac{1}{a} \right)$$

$$1.25.3 \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a} \left(x^2 - \frac{2x}{a} + \frac{2}{a^2} \right)$$

$$1.25.4 \quad \int x^n e^{ax} dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx \\ = \frac{e^{ax}}{a} \left(x^n - \frac{nx^{n-1}}{a} + \frac{n(n-1)x^{n-2}}{a^2} - \dots + \frac{(-1)^n n!}{a^n} \right) \quad \text{si } n = \text{entero positivo.}$$

$$1.25.5 \quad \int \frac{e^{ax}}{x} dx = \ln x + \frac{ax}{1 \cdot 1!} + \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^3}{3 \cdot 3!} + \dots$$

$$1.25.6 \quad \int \frac{e^{ax}}{x^n} dx = \frac{-e^{ax}}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{e^{ax}}{x^{n-1}} dx$$

$$1.25.7 \quad \int \frac{dx}{p + qe^{ax}} = \frac{x}{p} - \frac{1}{ap} \ln(p + qe^{ax})$$

$$1.25.8 \quad \int \frac{dx}{(p + qe^{ax})^2} = \frac{x}{p^2} + \frac{1}{ap(p + qe^{ax})} - \frac{1}{ap^2} \ln(p + qe^{ax})$$

$$1.25.9 \quad \int \frac{dx}{pe^{ax} + qe^{-ax}} = \begin{cases} \frac{1}{a\sqrt{pq}} \operatorname{tg}^{-1} \left(\sqrt{\frac{p}{q}} e^{ax} \right) \\ \frac{1}{2a\sqrt{-pq}} \ln \left(\frac{e^{ax} - \sqrt{-q/p}}{e^{ax} + \sqrt{-q/p}} \right) \end{cases}$$

$$1.25.10 \quad \int e^{ax} \sin bx \, dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2}$$

$$1.25.11 \quad \int e^{ax} \cos bx \, dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$$

$$1.25.12 \quad \int x e^{ax} \sin bx \, dx = \frac{x e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} - \frac{e^{ax}\{(a^2 - b^2) \sin bx - 2ab \cos bx\}}{(a^2 + b^2)^2}$$

$$1.25.13 \quad \int x e^{ax} \cos bx \, dx = \frac{x e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} - \frac{e^{ax}\{(a^2 - b^2) \cos bx - 2ab \sin bx\}}{(a^2 + b^2)^2}$$

$$1.25.14 \quad \int e^{ax} \ln x \, dx = \frac{e^{ax} \ln x}{a} - \frac{1}{a} \int \frac{e^{ax}}{x} \, dx$$

$$1.25.15 \quad \int e^{ax} \sin^n bx \, dx = \frac{e^{ax} \sin^{n-1} bx}{a^2 + n^2 b^2} (a \sin bx - nb \cos bx) + \frac{n(n-1)b^2}{a^2 + n^2 b^2} \int e^{ax} \sin^{n-2} bx \, dx$$

$$1.25.16 \quad \int e^{ax} \cos^n bx \, dx = \frac{e^{ax} \cos^{n-1} bx}{a^2 + n^2 b^2} (a \cos bx + nb \sin bx) + \frac{n(n-1)b^2}{a^2 + n^2 b^2} \int e^{ax} \cos^{n-2} bx \, dx$$

(26)

INTEGRALES QUE CONTIENEN $\ln x$

$$1.26.1 \quad \int \ln x \, dx = x \ln x - x$$

$$1.26.2 \quad \int x \ln x \, dx = \frac{x^2}{2} (\ln x - \frac{1}{2})$$

$$1.26.3 \quad \int x^m \ln x \, dx = \frac{x^{m+1}}{m+1} \left(\ln x - \frac{1}{m+1} \right) \quad [\text{Si } m = -1, \text{ véase 1.26.4}].$$

$$1.26.4 \quad \int \frac{\ln x}{x} \, dx = \frac{1}{2} \ln^2 x$$

$$1.26.5 \quad \int \frac{\ln x}{x^2} \, dx = -\frac{\ln x}{x} - \frac{1}{x}$$

$$1.26.6 \quad \int \ln^2 x \, dx = x \ln^2 x - 2x \ln x + 2x$$

$$1.26.7 \quad \int \frac{\ln^n x \, dx}{x} = \frac{\ln^{n+1} x}{n+1} \quad [\text{Si } n = -1, \text{ véase 1.26.8}].$$

$$1.26.8 \quad \int \frac{dx}{x \ln x} = \ln(\ln x)$$

$$1.26.9 \quad \int \frac{dx}{\ln x} = \ln(\ln x) + \ln x + \frac{\ln^2 x}{2 \cdot 2!} + \frac{\ln^3 x}{3 \cdot 3!} + \dots$$

$$1.26.10 \quad \int \frac{x^m dx}{\ln x} = \ln(\ln x) + (m+1) \ln x + \frac{(m+1)^2 \ln^2 x}{2 \cdot 2!} + \frac{(m+1)^3 \ln^3 x}{3 \cdot 3!} + \dots$$

$$1.26.11 \quad \int \ln^n x dx = x \ln^n x - n \int \ln^{n-1} x dx$$

$$1.26.12 \quad \int x^m \ln^n x dx = \frac{x^{m+1} \ln^n x}{m+1} - \frac{n}{m+1} \int x^m \ln^{n-1} x dx \quad [\text{Si } m = -1, \text{ véase 1.26.7}].$$

$$1.26.13 \quad \int \ln(x^2 + a^2) dx = x \ln(x^2 + a^2) - 2x + 2a \operatorname{tg}^{-1} \frac{x}{a}$$

$$1.26.14 \quad \int \ln(x^2 - a^2) dx = x \ln(x^2 - a^2) - 2x + a \ln \left(\frac{x+a}{x-a} \right)$$

$$1.26.15 \quad \int x^m \ln(x^2 \pm a^2) dx = \frac{x^{m+1} \ln(x^2 \pm a^2)}{m+1} - \frac{2}{m+1} \int \frac{x^{m+2}}{x^2 \pm a^2} dx$$

(27) INTEGRALES QUE CONTIENEN $\operatorname{sh} ax$

$$1.27.1 \quad \int \operatorname{sh} ax dx = \frac{\operatorname{ch} ax}{a}$$

$$1.27.2 \quad \int x \operatorname{sh} ax dx = \frac{x \operatorname{ch} ax}{a} - \frac{\operatorname{sh} ax}{a^2}$$

$$1.27.3 \quad \int x^2 \operatorname{sh} ax dx = \left(\frac{x^2}{a} + \frac{2}{a^3} \right) \operatorname{ch} ax - \frac{2x}{a^2} \operatorname{sh} ax$$

$$1.27.4 \quad \int \frac{\operatorname{sh} ax}{x} dx = ax + \frac{(ax)^3}{3 \cdot 3!} + \frac{(ax)^5}{5 \cdot 5!} + \dots$$

$$1.27.5 \quad \int \frac{\operatorname{sh} ax}{x^2} dx = -\frac{\operatorname{sh} ax}{x} + a \int \frac{\operatorname{ch} ax}{x} dx \quad [\text{Véase 1.28.4}].$$

$$1.27.6 \quad \int \frac{dx}{\operatorname{sh} ax} = \frac{1}{a} \ln \operatorname{th} \frac{ax}{2}$$

$$1.27.7 \quad \int \frac{x dx}{\operatorname{sh} ax} = \frac{1}{a^2} \left\{ ax - \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} - \dots + \frac{2(-1)^n (2^{2n} - 1) B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.27.8 \quad \int \operatorname{sh}^2 ax \, dx = \frac{\operatorname{sh} ax \operatorname{ch} ax}{2a} - \frac{x}{2}$$

$$1.27.9 \quad \int x \operatorname{sh}^2 ax \, dx = \frac{x \operatorname{sh} 2ax}{4a} - \frac{\operatorname{ch} 2ax}{8a^2} - \frac{x^2}{4}$$

$$1.27.10 \quad \int \frac{dx}{\operatorname{sh}^2 ax} = -\frac{\operatorname{coth} ax}{a}$$

$$1.27.11 \quad \int \operatorname{sh} ax \operatorname{sh} px \, dx = \frac{\operatorname{sh} (a+p)x}{2(a+p)} - \frac{\operatorname{sh} (a-p)x}{2(a-p)}$$

Para $a = \pm p$, véase 1.27.8.

$$1.27.12 \quad \int x^m \operatorname{sh} ax \, dx = \frac{x^m \operatorname{ch} ax}{a} - \frac{m}{a} \int x^{m-1} \operatorname{ch} ax \, dx \quad [\text{Véase 1.28.12}].$$

$$1.27.13 \quad \int \operatorname{sh}^n ax \, dx = \frac{\operatorname{sh}^{n-1} ax \operatorname{ch} ax}{an} - \frac{n-1}{n} \int \operatorname{sh}^{n-2} ax \, dx$$

$$1.27.14 \quad \int \frac{\operatorname{sh} ax}{x^n} dx = \frac{-\operatorname{sh} ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\operatorname{ch} ax}{x^{n-1}} dx \quad [\text{Véase 1.28.14}].$$

$$1.27.15 \quad \int \frac{dx}{\operatorname{sh}^n ax} = \frac{-\operatorname{ch} ax}{a(n-1)\operatorname{sh}^{n-1} ax} - \frac{n-2}{n-1} \int \frac{dx}{\operatorname{sh}^{n-2} ax}$$

$$1.27.16 \quad \int \frac{x \, dx}{\operatorname{sh}^n ax} = \frac{-x \operatorname{ch} ax}{a(n-1)\operatorname{sh}^{n-1} ax} - \frac{1}{a^2(n-1)(n-2)\operatorname{sh}^{n-2} ax} - \frac{n-2}{n-1} \int \frac{x \, dx}{\operatorname{sh}^{n-2} ax}$$

(28)

INTEGRALES QUE CONTIENEN $\operatorname{ch} ax$

$$1.28.1 \quad \int \operatorname{ch} ax \, dx = \frac{\operatorname{sh} ax}{a}$$

$$1.28.2 \quad \int x \operatorname{ch} ax \, dx = \frac{x \operatorname{sh} ax}{a} - \frac{\operatorname{ch} ax}{a^2}$$

$$1.28.3 \quad \int x^2 \operatorname{ch} ax \, dx = -\frac{2x \operatorname{ch} ax}{a^2} + \left(\frac{x^2}{a} + \frac{2}{a^3} \right) \operatorname{sh} ax$$

$$1.28.4 \quad \int \frac{\operatorname{ch} ax}{x} dx = \ln x + \frac{(ax)^2}{2 \cdot 2!} + \frac{(ax)^4}{4 \cdot 4!} + \frac{(ax)^6}{6 \cdot 6!} + \dots$$

$$1.28.5 \quad \int \frac{\operatorname{ch} ax}{x^2} dx = -\frac{\operatorname{ch} ax}{x} + a \int \frac{\operatorname{sh} ax}{x} dx \quad [\text{Véase 1.27.4}].$$

$$1.28.6 \quad \int \frac{dx}{\operatorname{ch} ax} = \frac{2}{a} \operatorname{tg}^{-1} e^{ax}$$

$$1.28.7 \quad \int \frac{x dx}{\operatorname{ch} ax} = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} - \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots + \frac{(-1)^n E_n(ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$1.28.8 \quad \int \operatorname{ch}^2 ax dx = \frac{x}{2} + \frac{\operatorname{sh} ax \operatorname{ch} ax}{2a}$$

$$1.28.9 \quad \int x \operatorname{ch}^2 ax dx = \frac{x^2}{4} + \frac{x \operatorname{sh} 2ax}{4a} - \frac{\operatorname{ch} 2ax}{8a^2}$$

$$1.28.10 \quad \int \frac{dx}{\operatorname{ch}^2 ax} = \frac{\operatorname{th} ax}{a}$$

$$1.28.11 \quad \int \operatorname{ch} ax \operatorname{ch} px dx = \frac{\operatorname{sh}(a-p)x}{2(a-p)} + \frac{\operatorname{sh}(a+p)x}{2(a+p)}$$

$$1.28.12 \quad \int x^m \operatorname{ch} ax dx = \frac{x^m \operatorname{sh} ax}{a} - \frac{m}{a} \int x^{m-1} \operatorname{sh} ax dx \quad [\text{Véase 1.27.12}].$$

$$1.28.13 \quad \int \operatorname{ch}^n ax dx = \frac{\operatorname{ch}^{n-1} ax \operatorname{sh} ax}{an} + \frac{n-1}{n} \int \operatorname{ch}^{n-2} ax dx$$

$$1.28.14 \quad \int \frac{\operatorname{ch} ax}{x^n} dx = \frac{-\operatorname{ch} ax}{(n-1)x^{n-1}} + \frac{a}{n-1} \int \frac{\operatorname{sh} ax}{x^{n-1}} dx \quad [\text{Véase 1.27.14}].$$

$$1.28.15 \quad \int \frac{dx}{\operatorname{ch}^n ax} = \frac{\operatorname{sh} ax}{a(n-1) \operatorname{ch}^{n-1} ax} + \frac{n-2}{n-1} \int \frac{dx}{\operatorname{ch}^{n-2} ax}$$

$$1.28.16 \quad \int \frac{x dx}{\operatorname{ch}^n ax} = \frac{x \operatorname{sh} ax}{a(n-1) \operatorname{ch}^{n-1} ax} + \frac{1}{(n-1)(n-2)a^2 \operatorname{ch}^{n-2} ax} + \frac{n-2}{n-1} \int \frac{x dx}{\operatorname{ch}^{n-2} ax}$$

(29) INTEGRALES QUE CONTIENEN $\operatorname{sh} ax$ Y $\operatorname{ch} ax$

$$1.29.1 \quad \int \operatorname{sh} ax \operatorname{ch} ax dx = \frac{\operatorname{sh}^2 ax}{2a}$$

$$1.29.2 \quad \int \operatorname{sh} px \operatorname{ch} qx dx = \frac{\operatorname{ch}(p+q)x}{2(p+q)} + \frac{\operatorname{ch}(p-q)x}{2(p-q)}$$

$$1.29.3 \quad \int \operatorname{sh}^2 ax \operatorname{ch}^2 ax dx = \frac{\operatorname{sh} 4ax}{32a} - \frac{x}{8}$$

$$1.29.4 \quad \int \frac{dx}{\operatorname{sh} ax \operatorname{ch} ax} = \frac{1}{a} \ln \operatorname{th} ax$$

$$1.29.5 \quad \int \frac{dx}{\operatorname{sh}^2 ax \operatorname{ch}^2 ax} = -\frac{2 \coth 2ax}{a}$$

$$1.29.6 \quad \int \frac{\operatorname{sh}^2 ax}{\operatorname{ch} ax} dx = \frac{\operatorname{sh} ax}{a} - \frac{1}{a} \operatorname{tg}^{-1} \operatorname{sh} ax$$

$$1.29.7 \quad \int \frac{\operatorname{ch}^2 ax}{\operatorname{sh} ax} dx = \frac{\operatorname{ch} ax}{a} + \frac{1}{a} \ln \operatorname{th} \frac{ax}{2}$$

(30)**INTEGRALES QUE CONTIENEN $\operatorname{th} ax$**

$$1.30.1 \quad \int \operatorname{th} ax dx = \frac{1}{a} \ln \operatorname{ch} ax$$

$$1.30.2 \quad \int \operatorname{th}^2 ax dx = x - \frac{\operatorname{th} ax}{a}$$

$$1.30.3 \quad \int \operatorname{th}^3 ax dx = \frac{1}{a} \ln \operatorname{ch} ax - \frac{\operatorname{th}^2 ax}{2a}$$

$$1.30.4 \quad \int x \operatorname{th} ax dx = \frac{1}{a^2} \left\{ \frac{(ax)^3}{3} - \frac{(ax)^5}{15} + \frac{2(ax)^7}{105} - \dots \frac{(-1)^{n-1} 2^{2n} (2^{2n} - 1) B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.30.5 \quad \int x \operatorname{th}^2 ax dx = \frac{x^2}{2} - \frac{x \operatorname{th} ax}{a} + \frac{1}{a^2} \ln \operatorname{ch} ax$$

$$1.30.6 \quad \int \frac{\operatorname{th} ax}{x} dx = ax - \frac{(ax)^3}{9} + \frac{2(ax)^5}{75} - \dots \frac{(-1)^{n-1} 2^{2n} (2^{2n} - 1) B_n (ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$1.30.7 \quad \int \frac{dx}{p + q \operatorname{th} ax} = \frac{px}{p^2 - q^2} - \frac{q}{a(p^2 - q^2)} \ln (q \operatorname{sh} ax + p \operatorname{ch} ax)$$

$$1.30.8 \quad \int \operatorname{th}^n ax dx = \frac{-\operatorname{th}^{n-1} ax}{a(n-1)} + \int \operatorname{th}^{n-2} ax dx$$

(31)**INTEGRALES QUE CONTIENEN $\operatorname{coth} ax$**

$$1.31.1 \quad \int \operatorname{coth} ax dx = \frac{1}{a} \ln \operatorname{sh} ax$$

$$1.31.2 \quad \int \operatorname{coth}^2 ax dx = x - \frac{\operatorname{coth} ax}{a}$$

$$1.31.3 \quad \int \operatorname{coth}^3 ax dx = \frac{1}{a} \ln \operatorname{sh} ax - \frac{\operatorname{coth}^2 ax}{2a}$$

$$1.31.4 \quad \int x \operatorname{coth} ax dx = \frac{1}{a^2} \left\{ ax + \frac{(ax)^3}{9} - \frac{(ax)^5}{225} + \dots \frac{(-1)^{n-1} 2^{2n} B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.31.5 \quad \int x \operatorname{coth}^2 ax dx = \frac{x^2}{2} - \frac{x \operatorname{coth} ax}{a} + \frac{1}{a^2} \ln \operatorname{sh} ax$$

$$1.31.6 \quad \int \frac{\coth ax}{x} dx = -\frac{1}{ax} + \frac{ax}{3} - \frac{(ax)^3}{135} + \dots - \frac{(-1)^n 2^{2n} B_n (ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$1.31.7 \quad \int \frac{dx}{p + q \coth ax} = \frac{px}{p^2 - q^2} - \frac{q}{a(p^2 - q^2)} \ln(p \operatorname{sh} ax + q \operatorname{ch} ax)$$

$$1.31.8 \quad \int \coth^n ax dx = -\frac{\coth^{n-1} ax}{a(n-1)} + \int \coth^{n-2} ax dx$$

(32) INTEGRALES QUE CONTIENEN $\operatorname{sech} ax$

$$1.32.1 \quad \int \operatorname{sech} ax dx = \frac{2}{a} \operatorname{tg}^{-1} e^{ax}$$

$$1.32.2 \quad \int \operatorname{sech}^2 ax dx = \frac{\operatorname{th} ax}{a}$$

$$1.32.3 \quad \int \operatorname{sech}^3 ax dx = \frac{\operatorname{sech} ax \operatorname{th} ax}{2a} + \frac{1}{2a} \operatorname{tg}^{-1} \operatorname{sh} ax$$

$$1.32.4 \quad \int x \operatorname{sech} ax dx = \frac{1}{a^2} \left\{ \frac{(ax)^2}{2} - \frac{(ax)^4}{8} + \frac{5(ax)^6}{144} + \dots - \frac{(-1)^n E_n (ax)^{2n+2}}{(2n+2)(2n)!} + \dots \right\}$$

$$1.32.5 \quad \int x \operatorname{sech}^2 ax dx = \frac{x \operatorname{th} ax}{a} - \frac{1}{a^2} \ln \operatorname{ch} ax$$

$$1.32.6 \quad \int \frac{\operatorname{sech} ax}{x} dx = \ln x - \frac{(ax)^2}{4} + \frac{5(ax)^4}{96} - \frac{61(ax)^6}{4320} + \dots - \frac{(-1)^n E_n (ax)^{2n}}{2n(2n)!} + \dots$$

$$1.32.7 \quad \int \operatorname{sech}^n ax dx = \frac{\operatorname{sech}^{n-2} ax \operatorname{th} ax}{a(n-1)} + \frac{n-2}{n-1} \int \operatorname{sech}^{n-2} ax dx$$

(33) INTEGRALES QUE CONTIENEN $\operatorname{csch} ax$

$$1.33.1 \quad \int \operatorname{csch} ax dx = \frac{1}{a} \ln \operatorname{th} \frac{ax}{2}$$

$$1.33.2 \quad \int \operatorname{csch}^2 ax dx = -\frac{\coth ax}{a}$$

$$1.33.3 \quad \int \operatorname{csch}^3 ax dx = -\frac{\operatorname{csch} ax \coth ax}{2a} - \frac{1}{2a} \ln \operatorname{th} \frac{ax}{2}$$

$$1.33.4 \quad \int x \operatorname{csch} ax dx = \frac{1}{a^2} \left\{ ax - \frac{(ax)^3}{18} + \frac{7(ax)^5}{1800} + \dots + \frac{2(-1)^n (2^{2n-1} - 1) B_n (ax)^{2n+1}}{(2n+1)!} + \dots \right\}$$

$$1.33.5 \quad \int x \operatorname{csch}^2 ax \, dx = -\frac{x \coth ax}{a} + \frac{1}{a^2} \ln \operatorname{sh} ax$$

$$1.33.6 \quad \int \frac{\operatorname{csch} ax}{x} dx = -\frac{1}{ax} - \frac{ax}{6} + \frac{7(ax)^3}{1080} + \dots \frac{(-1)^n 2(2^{2n-1} - 1) B_n (ax)^{2n-1}}{(2n-1)(2n)!} + \dots$$

$$1.33.7 \quad \int \operatorname{csch}^n ax \, dx = \frac{-\operatorname{csch}^{n-2} ax \coth ax}{a(n-1)} - \frac{n-2}{n-1} \int \operatorname{csch}^{n-2} ax \, dx$$

(34) INTEGRALES QUE CONTIENEN FUNCIONES HIPERBÓLICAS INVERSAS

$$1.34.1 \quad \int \operatorname{sh}^{-1} \frac{x}{a} dx = x \operatorname{sh}^{-1} \frac{x}{a} - \sqrt{x^2 + a^2}$$

$$1.34.2 \quad \int x \operatorname{sh}^{-1} \frac{x}{a} dx = \left(\frac{x^2}{2} + \frac{a^2}{4} \right) \operatorname{sh}^{-1} \frac{x}{a} - \frac{x \sqrt{x^2 + a^2}}{4}$$

$$1.34.3 \quad \int \frac{\operatorname{sh}^{-1}(x/a)}{x} dx = \begin{cases} \frac{x}{a} - \frac{(x/a)^3}{2 \cdot 3 \cdot 3} + \frac{1 \cdot 3 (x/a)^5}{2 \cdot 4 \cdot 5 \cdot 5} - \frac{1 \cdot 3 \cdot 5 (x/a)^7}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 7} + \dots & |x| < a \\ \frac{\ln^2(2x/a)}{2} - \frac{(a/x)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3 (a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} - \frac{1 \cdot 3 \cdot 5 (a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} + \dots & x > a \\ -\frac{\ln^2(-2x/a)}{2} + \frac{(a/x)^2}{2 \cdot 2 \cdot 2} - \frac{1 \cdot 3 (a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \frac{1 \cdot 3 \cdot 5 (a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} - \dots & x < -a \end{cases}$$

$$1.34.4 \quad \int \operatorname{ch}^{-1} \frac{x}{a} dx = \begin{cases} x \operatorname{ch}^{-1}(x/a) - \sqrt{x^2 - a^2}, & \operatorname{ch}^{-1}(x/a) > 0 \\ x \operatorname{ch}^{-1}(x/a) + \sqrt{x^2 - a^2}, & \operatorname{ch}^{-1}(x/a) < 0 \end{cases}$$

$$1.34.5 \quad \int x \operatorname{ch}^{-1} \frac{x}{a} dx = \begin{cases} \frac{1}{4}(2x^2 - a^2) \operatorname{ch}^{-1}(x/a) - \frac{1}{4}x \sqrt{x^2 - a^2}, & \operatorname{ch}^{-1}(x/a) > 0 \\ \frac{1}{4}(2x^2 - a^2) \operatorname{ch}^{-1}(x/a) + \frac{1}{4}x \sqrt{x^2 - a^2}, & \operatorname{ch}^{-1}(x/a) < 0 \end{cases}$$

$$1.34.6 \quad \int \frac{\operatorname{ch}^{-1}(x/a)}{x} dx = \pm \left[\frac{1}{2} \ln^2(2x/a) + \frac{(a/x)^2}{2 \cdot 2 \cdot 2} + \frac{1 \cdot 3 (a/x)^4}{2 \cdot 4 \cdot 4 \cdot 4} + \frac{1 \cdot 3 \cdot 5 (a/x)^6}{2 \cdot 4 \cdot 6 \cdot 6 \cdot 6} + \dots \right] \\ + \text{si } \operatorname{ch}^{-1}(x/a) > 0, \quad - \text{si } \operatorname{ch}^{-1}(x/a) < 0.$$

$$1.34.7 \quad \int \operatorname{th}^{-1} \frac{x}{a} dx = x \operatorname{th}^{-1} \frac{x}{a} + \frac{a}{2} \ln(a^2 - x^2)$$

$$1.34.8 \quad \int x \operatorname{th}^{-1} \frac{x}{a} dx = \frac{ax}{2} + \frac{1}{2}(x^2 - a^2) \operatorname{th}^{-1} \frac{x}{a}$$

$$1.34.9 \quad \int \frac{\operatorname{th}^{-1}(x/a)}{x} dx = \frac{x}{a} + \frac{(x/a)^3}{3^2} + \frac{(x/a)^5}{5^2} + \dots$$

$$1.34.10 \quad \int \operatorname{coth}^{-1} \frac{x}{a} dx = x \operatorname{coth}^{-1} x + \frac{a}{2} \ln(x^2 - a^2)$$

$$1.34.11 \quad \int x \coth^{-1} \frac{x}{a} dx = \frac{ax}{2} + \frac{1}{2}(x^2 - a^2) \coth^{-1} \frac{x}{a}$$

$$1.34.12 \quad \int \frac{\coth^{-1}(x/a)}{x} dx = - \left(\frac{a}{x} + \frac{(a/x)^3}{3^2} + \frac{(a/x)^5}{5^2} + \dots \right)$$

$$1.34.13 \quad \int \operatorname{sech}^{-1} \frac{x}{a} dx = \begin{cases} x \operatorname{sech}^{-1}(x/a) + a \operatorname{sen}^{-1}(x/a), & \operatorname{sech}^{-1}(x/a) > 0 \\ x \operatorname{sech}^{-1}(x/a) - a \operatorname{sen}^{-1}(x/a), & \operatorname{sech}^{-1}(x/a) < 0 \end{cases}$$

$$1.34.14 \quad \int \operatorname{csch}^{-1} \frac{x}{a} dx = x \operatorname{csch}^{-1} \frac{x}{a} \pm a \operatorname{sh}^{-1} \frac{x}{a} \quad [+ \text{ si } x > 0, \quad - \text{ si } x < 0].$$

$$1.34.15 \quad \int x^m \operatorname{sh}^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \operatorname{sh}^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 + a^2}} dx$$

$$1.34.16 \quad \int x^m \operatorname{ch}^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1}}{m+1} \operatorname{ch}^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 - a^2}} dx & \operatorname{ch}^{-1}(x/a) > 0 \\ \frac{x^{m+1}}{m+1} \operatorname{ch}^{-1} \frac{x}{a} + \frac{1}{m+1} \int \frac{x^{m+1}}{\sqrt{x^2 - a^2}} dx & \operatorname{ch}^{-1}(x/a) < 0 \end{cases}$$

$$1.34.17 \quad \int x^m \operatorname{th}^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \operatorname{th}^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^{m+1}}{a^2 - x^2} dx$$

$$1.34.18 \quad \int x^m \operatorname{coth}^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \operatorname{coth}^{-1} \frac{x}{a} - \frac{1}{m+1} \int \frac{x^{m+1}}{a^2 - x^2} dx$$

$$1.34.19 \quad \int x^m \operatorname{sech}^{-1} \frac{x}{a} dx = \begin{cases} \frac{x^{m+1}}{m+1} \operatorname{sech}^{-1} \frac{x}{a} + \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{a^2 - x^2}} & \operatorname{sech}^{-1}(x/a) > 0 \\ \frac{x^{m+1}}{m+1} \operatorname{sech}^{-1} \frac{x}{a} - \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{a^2 - x^2}} & \operatorname{sech}^{-1}(x/a) < 0 \end{cases}$$

$$1.34.20 \quad \int x^m \operatorname{csch}^{-1} \frac{x}{a} dx = \frac{x^{m+1}}{m+1} \operatorname{csch}^{-1} \frac{x}{a} \pm \frac{a}{m+1} \int \frac{x^m dx}{\sqrt{x^2 + a^2}} \quad [+ \text{ si } x > 0, \quad - \text{ si } x < 0].$$